



SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY!

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Measurement of Health and Disease (Epidemiology) For Pharmacy Students BY

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1. Introduction to Epidemiology

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▣ Contents

- Definition of epidemiology
- History of Epidemiology
- Fundamental assumptions of Epidemiology
- Purpose of Epidemiology
- Scope of Epidemiology
- Epidemiological Transitions
- Disease causation theories and Models
- Natural history of disease and levels of prevention
- Verities/Branches of Epidemiology

Introduction...

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- The word **epidemiology** comes from 3 **Greek** root words:

Epi: About/“on or upon,”

Demos: Human population/people.

Logia: Discipline/the study of.

- Many definitions have been proposed over time by different scholars.
- Epidemiology is an independent branch of medicine studying (disease etiology, spread and prevention) 1960 in Prague.

Oxford English Dictionary

- The branch of medical science which treats of epidemics- *old*
- The scientific study of the spread and control of diseases – *new*

Kuller LH: Am J Epid 1991;134:1051

- Epidemiology is the study of "epidemics" and their prevention

Introduction...

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Anderson G, quoted in Rothman KJ: Modern Epidemiology

- Epidemiology: the study of the occurrence of illness

Last JM: A Dictionary of Epidemiology

- The study of the distribution and determinants of health related states and events in populations and the application of this study to control of health problems.
- The following definition captures the underlying principles and the public health spirit of epidemiology.
- Its definition has CDC, Wikipedia, Epidemiologist, etc. origins.
- *Health* is derived from the Old English “*Hal*”, meaning whole, sound in wind and limb.

Introduction...

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“Is the **study** of the frequency, distribution and determinants of health related states or events in specified population and the application of this study to the promotion of health and prevention and control of health problems.”

Or

- “Is a science which studies the distribution and determinants of health and health related events in populations, and the application of this knowledge for the promotion of health, prevention and control of diseases.” ***Last, Dictionary of Epidemiology.***

What are Key words in the definition are?

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- ❖ **Study:** Epidemiology is a scientific discipline, sometimes called “**the basic science of public health.**” It has, at its foundation, sound methods of scientific inquiry.
 - ❖ It approaches problems systematically and quantitatively. **Epidemiology is a science**
- ❖ **Frequency**
 - ▢ Studies the number of times a disease occurs
 - ▢ Reminds us that epidemiology is a **quantitative science** i.e. **PPT**.
 - ▢ We measure frequency of death using mortality rates, which quantify occurrence of death and morbidity rates, which quantify occurrence of illness.
- ❖ **Human Population**
 - ▢ Epidemiology focuses on populations rather than individuals. It diagnose and treat communities or populations.
 - ▢ **Epidemiology is a basic science of public health**
 - ▢ Characterize a basic difference in approach between epidemiology and clinical medicine.
 - In **EPIDEMIOLOGY**, we study effects of diseases on populations, while in **CLINICAL MEDICINE** we study effects of diseases on individuals.
- ▢ So the epidemiologist’s “patient” is the **?**

Key word definitions...

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❖ Distribution

- ▣ Refers not only geographic distribution of diseases but also, to its distribution in time (**seasonal fluctuation**), according to the type of **persons** most or least affected (age group, occupation).
- ▣ **who? Where? and when? Of occurrence** of health and health related events
- ▣ **Epidemiology describes health events**

❖ Determinants(D)

- ▣ Factor which may be event, characteristic or any definable entity that brings about change in health and health related conditions.
- ▣ **It refers to “why diseases occur in certain places? In a certain period? or in a certain population groups?”**
- ▣ Refers to the ‘**why**’ and ‘**how**’ of disease occurrence
 - ***Cause, risk factors, mechanisms***
- ▣ D: Any factor that brings about change in a health condition or other defined characteristic. Single specified causes. A determinant makes a difference to a given outcome.
- ▣ D≠ Does not imply a deterministic philosophy of health.
- ▣ **Epidemiology analyze health events**

Key word definitions...

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Disease and other health related events

- Epidemiology is concerned not only with disease but events like birth, death, migration, etc.
 - Examples are; Injuries Vital events, birth, death, marriages, divorce , Health related behaviors (Sexual behavior, Smoking, Alcoholism, Drug abuse, Social factors.
- The central concern of epidemiology is **promotion and maintenance** of health through the prevention of diseases.
- **Epidemiology is a broader science**



Application- epidemiology is an **applied science**. The ultimate purpose of all epidemiological studies is the prevention and control of health problems.

❖ Epidemiology is a science and practice

- Epidemiologic investigation is problem-oriented and tends toward applied research
- Is a science and practice/art i.e. knowledge for action
- It is an applied science with direct practical applications
- An important aim in studying the frequency, distribution and determinants of diseases **is to identify effective diseases prevention and control strategies.**
- ◇ **Epidemiology is an applied science**

History of Epidemiology

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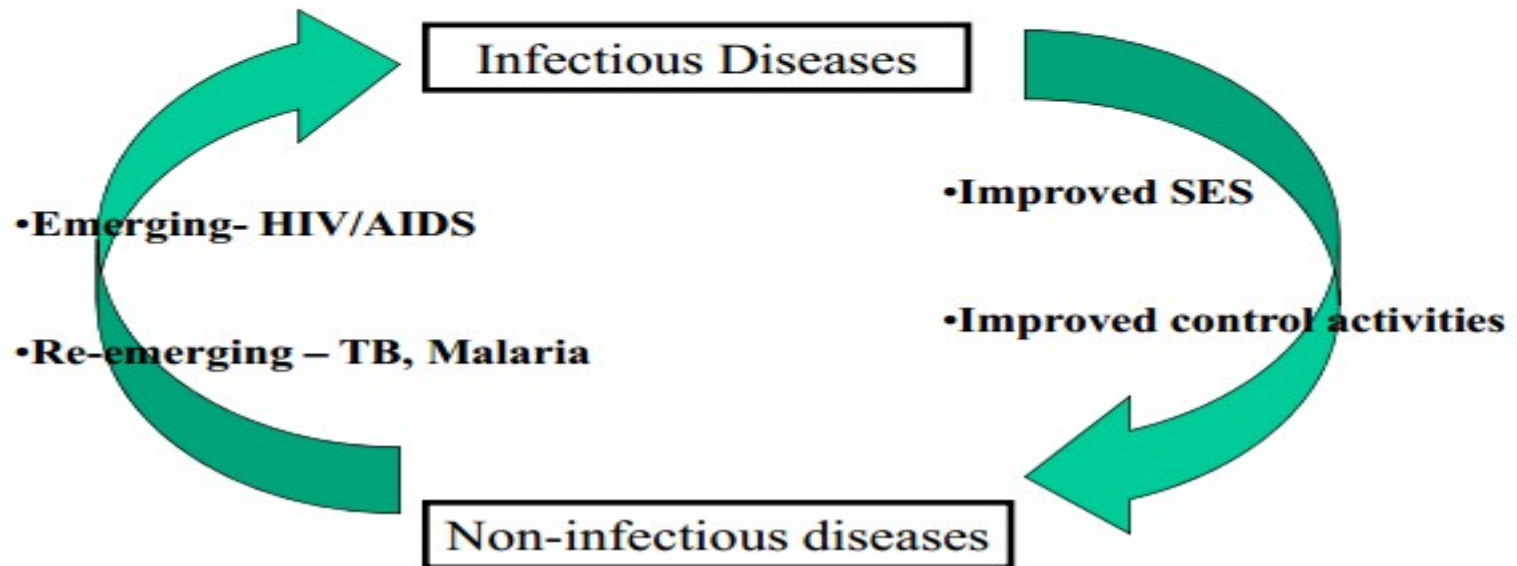
- ❑ Why teach the history of epidemiology?
- ❑ As history can be learnt so it can be taught. In this respect a basic consideration is that historical knowledge changes in time;
- ❑ Not only because of cumulative advances, as for all theoretical or empirical sciences,
- ❑ But also because the point of observation of the historical past moves continuously forward with time and the landscape that becomes visible to the eye of the observer changes in consequence.

History....

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Evolution of Epidemiology

1. Trends in Epidemiology

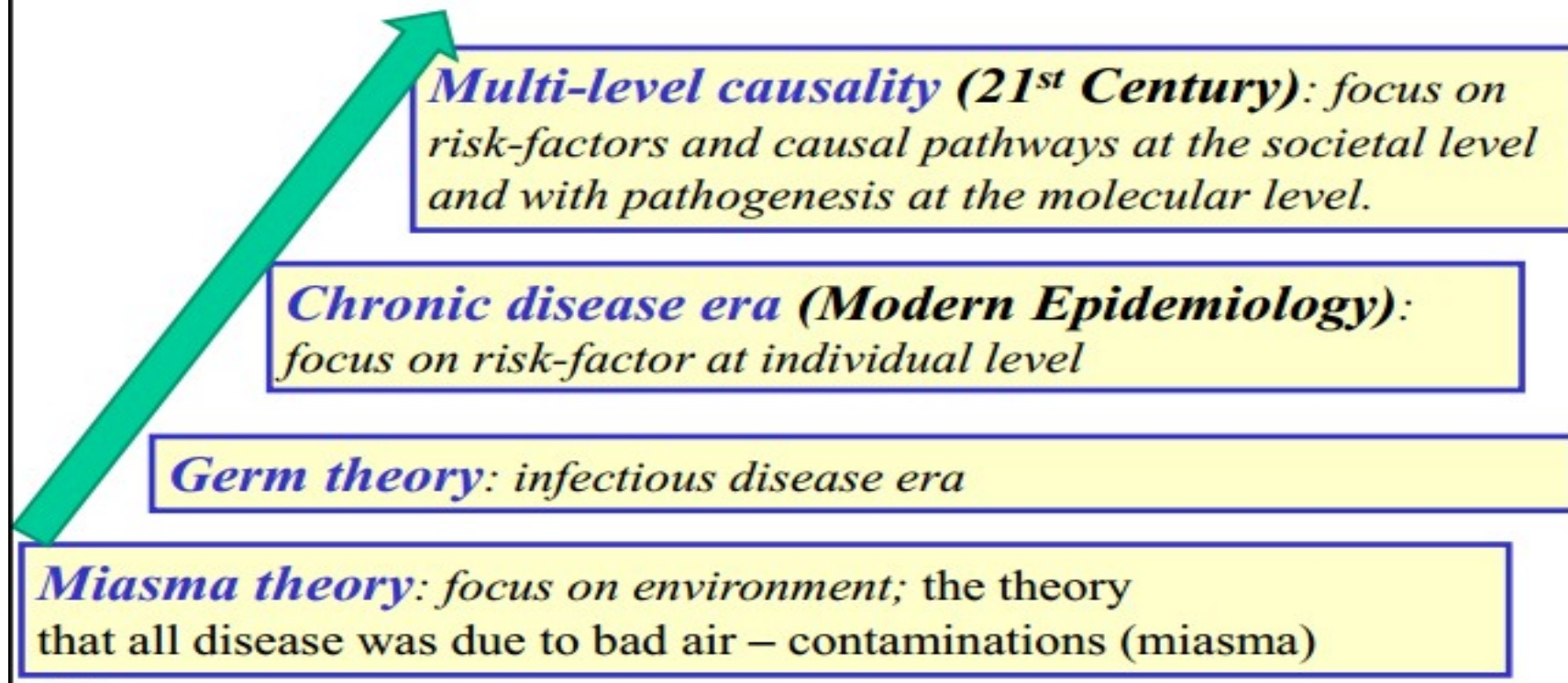


Epidemiological transition

History...

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2. Evolution in Epidemiology



History...

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There are many histories of Epidemiology, history of:

- ❑ Health and disease
- ❑ Ideas and concepts
- ❑ Methods
- ❑ Knowledge gained through these concepts and methods
- ❑ Teachers and students
- ❑ Organizations and actions

To review the **yesterday**, **today**, and **tomorrow** of epidemiology, better to consider three periods:

1. **Early Epidemiology, extending from the 15c BC to around 1830;**
2. **Classical Epidemiology, from around 1830-1940s;**
3. **New Epidemiology, from the 1940s to the present.**

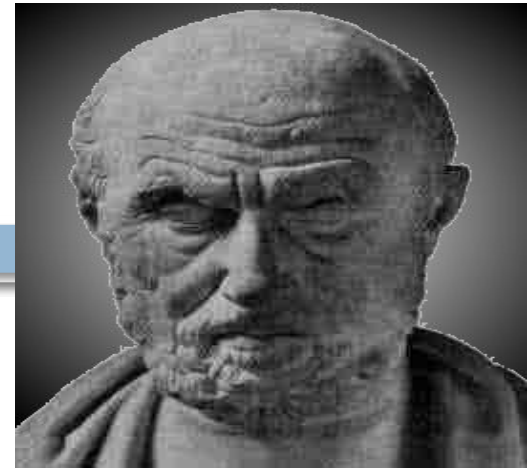
History...

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- The history of epidemiology originated as early as in years, before about ADs, and later during the medieval period, during the bubonic plague epidemics in Europe.
- The Renaissance was the time of Paracelsus (toxicologist) and John Graunt, pioneering compiler of vital statistics.
- During the eighteenth and nineteenth centuries, breakthroughs occurred in the development of a vaccination against smallpox and the formulation of epidemiologic methods.
- The period from the beginning of the twentieth century to the present has seen a rapid growth in epidemiology;
- Two of the achievements of this period were identification of smoking as a cause of cancer and eradication of smallpox.

Hippocrates (460-370 BC)

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□ *On Airs, Waters, and Places*

- ✓ Idea that disease might be associated with physical environment
- ✓ Environmental factors such as water quality and the air were implicated in the causation of diseases.
- ✓ The Greek physician Hippocrates is known as the father of medicine, and was the **first epidemiologist**.
- ✓ Hippocrates sought a logic to sickness.
- ✓ He is the **first** person known to have examined the relationships between the occurrence of disease and environmental influences.

Middle Ages (500-1450 AD) Black Death.

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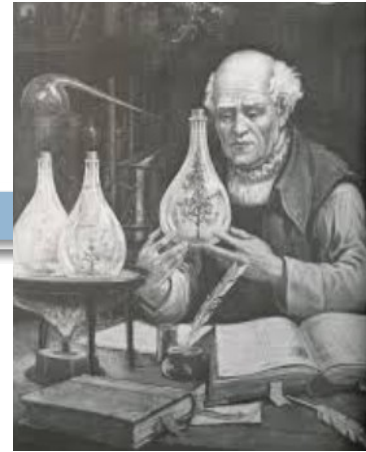


- It is of great significance for epidemiology
- It occurred between 1346 and 1352
- It is claimed that up to one-third of the population of Europe at the time (20 to 30 M of the 100 M) death.
- It was thought to be an epidemic of **bubonic plague(BP)**, a bacterial disease caused by *Yersinia Pestis* (formerly *Pasteurella pestis*).
- **BP** is a gram-negative, non-motile, rod-shaped, coccobacillus bacterium, without spores. It is a facultative anaerobic organism that can infect humans via the Oriental rat flea (*Xenopsylla cheopis*).

Paracelsus (1493-1541).

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- He was the founder of the field of *toxicology*,
- A discipline that is used to examine the toxic effects of chemicals found in environmental (the workplace).
- Paracelsus, advanced toxicology during the early sixteenth century.
- Among his contributions were several important concepts:
 - ▣ The dose response relationship

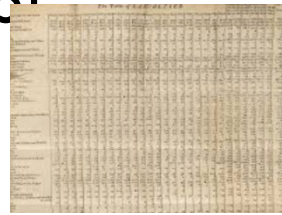


John Graunt (1620-1674).



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- He assessed a *recorded descriptive characteristics* of
 - ▣ Birth and death data,
 - ▣ Seasonal variations,
 - ▣ Infant mortality, and excess male over female mortality.
- Graunt is said to be the first to employ quantitative methods to describe population vital statistics by organizing mortality data in a mortality table.
- Because of his contributions to vital statistics, Graunt has been called the Columbus of statistics



Thomas Sydenham (1624-1689)

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- ✓ Recognized as a founder of,
 - ✓ Clinical medicine and epidemiology
- ✓ Emphasized detailed observations of,
 - ✓ Patients & accurate record keeping
- ✓ Proper founder of the natural history of diseases.



Bernardino Ramazzini (1633-1714)

20

- He is regarded as the founder of the field of
 - ▣ Occupational medicine.
- He created to elaborate descriptions of,
 - ▣ The manifestations of occupational diseases among,
 - ▣ Many different types of workers.
- His descriptions covered a excess of occupations, from miners to cleaners of toilets to fabric workers.
- The father of occupational medicine is also considered to be a inventor in the field of ergonomics, by pointing out the hazards associated with postures assumed in various occupations.



Sir Percival Pott (1714-1788)



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- He was a London surgeon, and is thought to be the first individual to describe an environmental cause of cancer.
- In 1775, he made the smart observation that chimney sweeps had a high incidence of scrotal cancer (in comparison with male workers in other occupations).
- He argued that chimney sweeps were prone to this malady as a consequence of their contact with soot.

Edward Jenner (1749-1823)



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- Jenner's findings was on vaccine that provided immunity to smallpox.
- He introduced the smallpox **vaccine** in 1796.
- The cowpox virus, produces a milder infection in humans than does the smallpox virus.
- He created a vaccine by using material from the arm of a dairy-maid, who had an active case of cowpox.
- The vaccine was injected into the arm of an 8 -year old boy, who was later exposed to smallpox and did not develop the disease.
- He concluding that the procedure was effective, and vaccinated other children including his own son

Edward J...

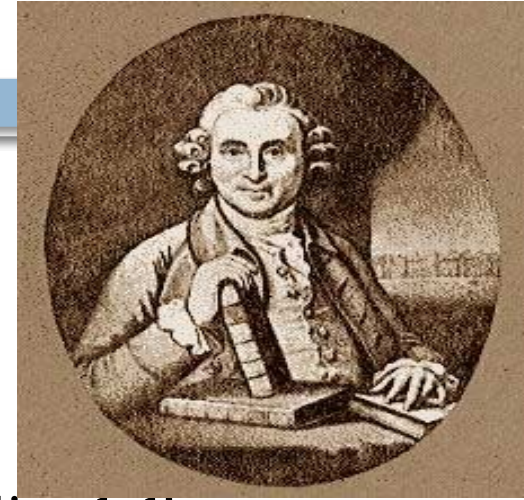
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- Jenner's work influenced many others, including Louis Pasteur who developed vaccines against rabies and other infectious diseases.



James Lind (1700's)

24



- Designed first experiments to use,
 - ▣ A concurrently treated control group
- In 1753, Lind, described an experiment
 - ▣ 12 sailors with scurvy were put on standardized diets allocated to one of six treatment groups for 14 days.
 - ▣ Those receiving oranges and lemons were much improved after six days.
- His study must be considered one of the first controlled clinical trials.

Ignas Semmelweis (1840's)

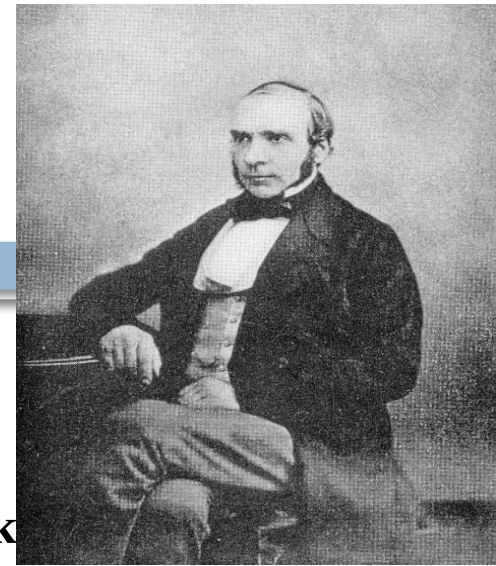
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- ❑ Pioneered hand washing to
 - Help prevent the spread of
 - Septic infections in mothers following birth
- ✓ Hand washing become the custom in assisting delivery



John Snow (1813-1858)

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- ✓ Widely considered to be the Father of epidemiology or

- ✓ **Modern ! Why?**

- ✓ The Broad Street **cholera outbreak**/Golden Square **outbreak**

- ✓ Careful mapping of cholera cases in East London during cholera epidemic of 1854

- ✓ A severe **outbreak** of **cholera** that occurred in 1854 near Broad Street (now Broadwick Street) in the Soho district of the City of Westminster, **London**, England, and occurred during the 1846–1860.

- ✓ The source was from three water companies: southwark and vauxhall water company, and lambeth

- ✓ Traced source to a single well on Broad Street that had been contaminated by sewage.

- ✓ Robert Koch (1843-1910) enjoyed worldwide fame, including acknowledgement of his **discovery** in 1882 of the tubercle bacillus that caused tuberculosis and in 1884 the **cholera** bacillus, **Vibrio cholerae**.

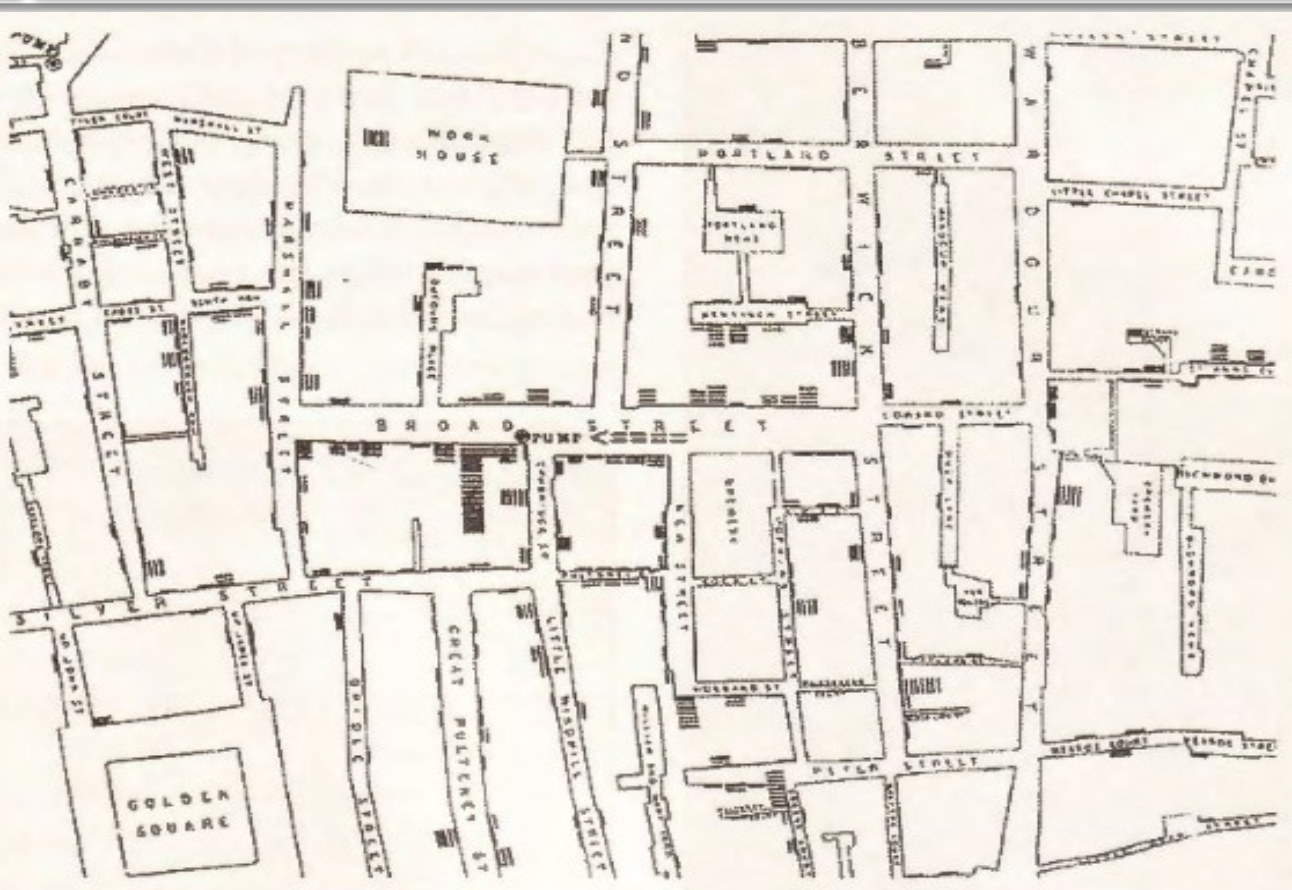
John S...

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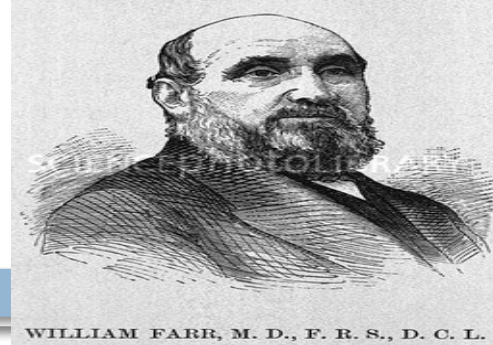
- Snow's contributions included :
 - Powers of observation and written expression
 - ▣ Application of epidemiologic methods
- Mapping (spot maps)
 - ▣ Use of data tables to describe infectious disease outbreaks
- Participation in a natural experiment
- Recommendation of a public health measure to prevent disease (removal of the pump handle)

Great street Pumps in London...

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William Farr (1807-1883)

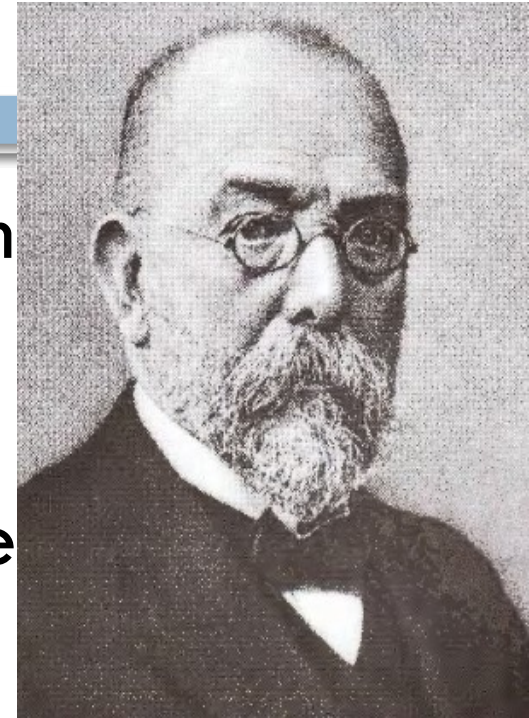


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- Farr's contributions to public health and epidemiology was the development of a more sophisticated system of data use and manipulation
- His creation of a vital statistics system, **role in** the formation of the International Classification of Diseases, and prominence in resolving the mode of communication of cholera in Victorian England were each seminal to modern **epidemiology**
- Farr used data such as census reports to study occupational mortality in England.
- In addition, he explored the possible linkage between mortality rates and population density.
- Founder of the modern concept of **surveillance &** medical statistics..

Robert Koch (1843-1910)

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- The German physician Robert Koch
 - ▣ Verified that a human disease,
 - was caused by a specific living organism.
- He isolated the bacteria that cause
 - ▣ Anthrax (*Bacillus anthracis*) , and
 - ▣ Cholera (*Vibrio cholerae*)
- One of his most famous contributions was identifying the cause of tuberculosis.

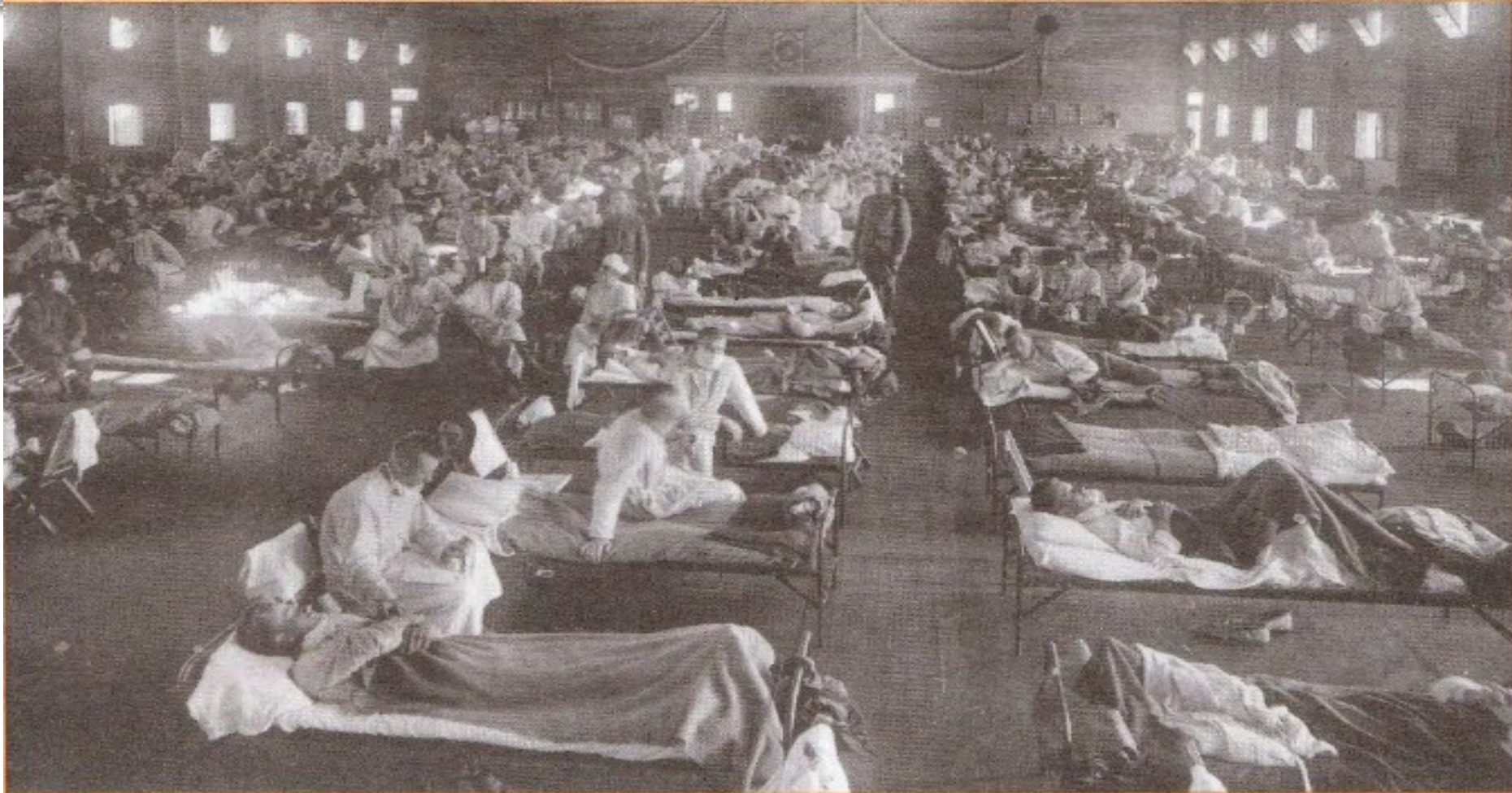
Early twentieth century (1900-1940) *pandemic influenza.*

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- Also known as the Spanish Flu, the pandemic erupted from 1918 to 1919 and killed 50 to 100 million persons globally.
- It was estimated to kill 33% of the world's population, (of the 1.5 billion)
- Instead of primarily attacking the young and the elderly, it heavily infected all including the healthy young adults.

Con't...

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Source: © National Museum of Health and Medicine, Armed Forces, Institute of Pathology, (NCI, 1903).

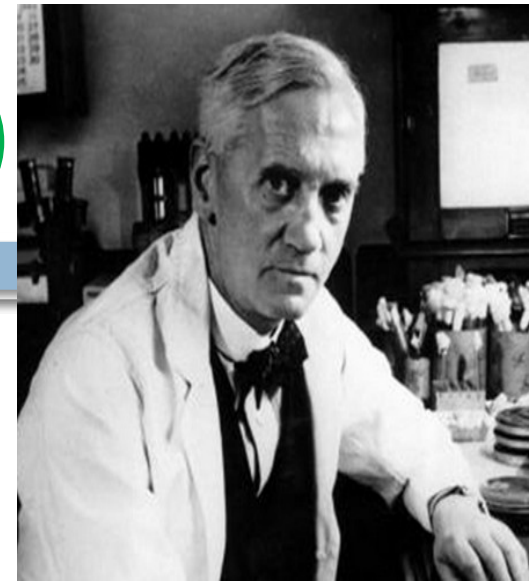
Testa B.

Epidemiology Lect. Note

09/25/2026

Alexander Fleming (1881-1955)

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- *Discovery of penicillin.*
- Scottish researcher discovered the antimicrobial properties of the mold *Penicillium notatum* in 1928.
- This innovation led to the development of the antibiotic penicillin, which became available toward the end of World War II.

The contemporary era (1940 to date)

34

- From the mid-twentieth century to the present epidemiology has made numerous contributions to society.
- These Innovations include:
 - ▣ Framingham Study. Begun in 1948, this pioneering research project is named for Framingham, Massachusetts. Initially, a random sample of 6,500 persons aged 30 to 59 years participated
 - ▣ Epidemic Intelligence Service. Alexander Langmuir was hired by the CDC as a chief epidemiologist. One of his contributions was the establishment in 1949 of the Epidemic Intelligence Service (EIS).

Con't...

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- ▣ Smoking and health. By the mid-twentieth century, a growing body of evidence suggested that cigarette smoking contributed to early mortality from lung cancer as well as other forms of morbidity and mortality.
- ▣ Smallpox eradication.
- ▣ As noted previously, **Jenner pioneered** development of a smallpox vaccine during the 1800s.
 - Smallpox is an incurable disease caused by a virus.
 - One form of the virus *variola major* produces a highly fatal infection in unvaccinated populations.
 - The last known naturally acquired case was reported in Somalia in 1977
- ◇ **Pioneer clinical, public or population epidemiologists contribution!**

Brief overview of current uses of epidemiology

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Among the principal uses of epidemiology are the following:

- **Historical use:** study the history of the health of populations
- **Community health use:** diagnose the health of the community
- **Health services use:** studying the function of health services
- **Risk assessment use:** estimating individuals' risks of disease, accident, or defect
- **Disease causality use:** search for the causes of health and disease
- ◇ **Practice of epidemiology**

History of clinical epidemiology

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- Is a subfield of epidemiology specifically focused on issues relevant to clinical medicine.
- The term **CE** first introduced by John Rodman Paul (American Virologist) in 1938.
- **CE** is the study of the patterns, causes, and effects of health and disease in patient populations and the relationships between exposures or treatments and health outcomes.
- **Lind and Louis** are two notable workers who used experimentation to assess the effect of a treatment on a disease.
- In 1753, Lind described an experiment
 - ▣ **12 sailors with scurvy** were put on standardized diets allocated to one of six treatment groups for 14 days.
 - ▣ Those **receiving oranges and lemons** were much improved after six days.
- His study must be considered one of the first **controlled clinical trials**.

Con't...

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- In 1834, Louis articulated guidance to follow regarding study design.
 - ▣ The number of subjects required (sample size)
 - ▣ The need to observe disease outcome on treated and controlled groups; (**end point**)
 - ▣ The need to define precisely entry criteria before the experiment;(inclusion criteria)

Con't

39

- The first randomized controlled trial was undertaken later, after 1940s by the Medical Research Council.
- Trials of **streptomycin** in the treatment TB (1948)
- Trials of **antihistamines** for the treatment of the common cold (1950) double-blind, placebo-controlled trial.
- In 1950/57, Cochran, W. G., & Cox, G. M. (1957) published an important textbook on experimental designs.
- This book was clearly and simply described the major statistical consideration relevant for experimental studies.

Con't

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- This book was clearly and simply described the major statistical consideration relevant for experimental studies including statistical inference, randomization, methods of increasing accuracy, replication of test of significance, statistical analysis of results, and various classes of experimental designs.



WILLIAM G. COCHRAN
William Cochran

Tesfa B.

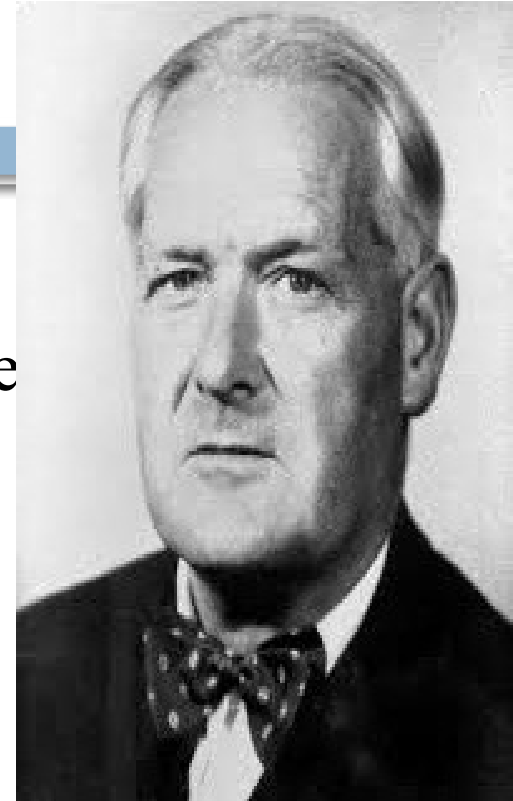


Sir Percy Cox

Bradford-Hill (1897 –1991)

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- He was also an important force in
 - ▣ Making the design of clinical trials more rigorous.
- He was an **English** epidemiologist and statistician who pioneered the **randomized clinical trial**
- *(Principles of medical statistics)*



Con't...

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- In 1959, Truelove(Dr.) summarized the current thinking on experimental design where he clearly described the essential elements of a **therapeutic trial on IBD/Colitis** as follows:
 1. The trial should be planned so that decisive answers can be given to important questions (**internal validity**)
 2. Patients should be selected for **inclusion** in the trial before it is known into which group they will go.
 3. After admission to the trial, patients should be allocated at **random** to one or other treatment group.

Con't...

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- 4. Systematic and pertinent observations should be made on patients so that relevant data are available for analysis at the **conclusion** of the trial.

- 4. When possible, **trials** should be so arranged that neither the physician nor the patient knows which treatment is being used - the so-called **double-blind system**.

Summary on history of Epidemiology

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Can be summarized in three periods

1. Early
 2. Classical
 3. New
- ◇ Overview of epidemiology

1. Early Epidemiology

45

- ❑ Ran for more than **two** millennia.
- ❑ Some key dates and contributions to the development of epidemiologic thinking and methods include;
 - ❑ **Hippocrates(c.470–c.400BC)** : He developed the medical approaches on various diseases by providing complete description of actual clinical cases.
 - ❑ The general dependence of health not on **magical** influences but on an identifiable array of natural external factors (identified seminal environments including air, water and places).

2. Classical epidemiology

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- With the advent of the industrial transformation 'crowd diseases' emerged which struck the populations amassed in the slums of the fast-growing centers of industrial development (western Europe).
- That creates decisive stimulus, observational field for epidemiology, which developed as the investigation facet of a vast public health movement
- In great Britain medical registration introduced 1801, and
- In 1838 William Farr (1807–1883) introduced a national system of recording causes of death (Life Table).
- John Snow (1813–1858), a contemporary of William Farr Who quoted as an example of a brilliant analytical investigation which can lead to the identification of a pathogenic agent and its elimination from the environment; Cholera (Asiatic cholera).

Classical...

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Snow's methods represent the modern foundation of
Epidemiological study

- Compared **cholera** rates by **neighborhood** –
 - **Ecological** studies
- Compared disease rates in exposed and unexposed persons (people who drew water from the Broad Street pump) –
 - **Cohort** studies
- Compared water source in infected and uninfected persons
 - **Case/control** studies.

Classical...

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- ❑ German scientist Rudolf Ludwig Carl Virchow (1821–1902)
- ❑ He is known as “The father of modern pathology”
- Founder of **microscopic** cellular pathology, but he also wrote and was very active in the field of **public health**, based on his belief that ‘**medicine** is a social science’.
- German scientist, **Robert Koch (1843–1910)**, in the wake of the fundamental discovery of **microorganisms** by **Louis Pasteur (1822–1895)**, not only to contribute to the discovery of the agents of several diseases (including the actual identification of the major disease **tuberculosis**),
- But also to formulate a set of criteria for establishing **causality** in epidemiological studies.

3. The new Epidemiology

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- ❑ Rise the period around **the Second World War**.
- ❑ Individual studies on cancer, non-rheumatic cardiovascular diseases, and psychiatric diseases, can be traced well back in time.
- ❑ The tobacco and health story developed (either of statistical nature and clinical).
- ❑ Insurance data showing that the expectation of life for smokers (by US mathematician **Raymond Pearl, 1938**)
- ❑ High frequency of smokers among the lung cancer patients coming to their hospital (by Ochsner and De Bakey, 1939)
- ❑ Causal between tobacco smoking and lung cancer (R. **Doll and Hill AB** ,1950)

The new Epidemiology...

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- **Epidemiology and clinical medicine**, which has been a constant feature in the history of the subject.
- As previously exemplified Ramazzini, a clinician, and Snow, a physician (pioneering anesthesia), had enlarged clinical observations by looking for causes of disease at the **population** level; **conversely**, Louis had brought the methods of population studies into the clinical domain to evaluate the effects of medical acts in patient populations.
- **Social Medicine** at Oxford, established at the time of the Second World War to investigate the influence of **social, genetic, environmental, and domestic factors** on the incidence of human **diseases and morbidity**.

Applications of Epidemiology

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1. Establishing **causation** of diseases

- Is an antecedent event, condition or characteristic that was necessary for the occurrence of the disease at the moment it occurred, given other conditions are fixed.

2. Studies (Elucidation) of the **natural history of diseases-NHD** (Careful following of the progression of disease outcome with out intervention)

- Is essential to make prognosis or the likely outcome of a patients illness.
- Gives basis for rational decision about therapy.
- Natural history of disease and autopsy data findings should be generalized only with great caution.
- **NHD**: The course of a disease from pathological onset or inception to resolution.

Application...

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3. Description of health status of populations

- Measure disease frequency- quantify disease
- Assess distribution of disease
 - ▣ Who is getting disease?
 - ▣ Where is disease occurring?
 - ▣ When is disease occurring?
 - Formulation of hypotheses concerning causal and preventive factors

Application...

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4. Identify determinants of diseases

- Purpose of epidemiologic study is the identification of determinants of disease, **whose manipulation could lead to prevention and control of diseases.**
- Hypothesis is developed after description of occurrence by person, place and time.
- Hypotheses are tested using epidemiologic studies
 - Determinants of health
 - Genetic factors
 - Lifestyle
 - Poverty and affluence
 - Health care facilities
 - Environment-physical, biological and psychological

Application...

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- ❖ Among the determinant, the most amenable for modification is the physical environment.
- ❖ Providing conducive physical environment implies:
 - **Safe water**
 - **Sanitary conveniences**
 - **Clean environment**
 - **Good food etc.**
- ❖ This involves a lot of spending and therefore it is often said that public health is “purchasable”.

Application...

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5. Evaluation of intervention programs

▣ Efficiency, effectiveness, feasibility and impact

- Any program designed to prevent and control a disease must be accompanied by methods for assessing whether the measures are effective in reducing the frequency of the disease.

6. Classification of diseases

▣ e.g. based on route of transmission

- Gonorrhea- as STD
- Shigellosis- under diarrhea diseases based on its clinical manifestation of diarrhea and as fecal- oral route of transmission.

Scope of Epidemiology

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- ❖ Originally, epidemiology was concerned with **epidemics** of communicable diseases, **lately**; epidemiology was extended to **endemic** communicable infectious and non-communicable diseases.
- ❖ More recently, epidemiologic methods have been applied to **chronic diseases, injuries, birth defects, maternal and child health, occupational health and environmental health.**
- ❖ Now, even health behaviors, such as **health care seeking, safety practices, violence and hygienic practices** are valid subjects for epidemiologic investigation.

Scope...

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- ❖ It is used in laboratory sciences, clinical medicine and public health
- ❖ Its scope in public health ranges from *routine surveillance to research strategies for the testing of hypotheses* about causes, measurement of health and disease risks and evaluations of preventive, diagnostic and therapeutic programmes and technologies.
- ❖ Epidemiology is also a collection of *applied/sisterly disciplines, i.e., every disease entity has its own epidemiology (infectious, cardio-vascular, cancer, etc.)*.

Scope of Epidemiology

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Scope of epidemiology

- Epidemics + communicable diseases
- Non-communicable diseases.
- At **present** epidemiologic methods are being applied to:
 - Infectious and non infectious diseases
 - Injuries and accidents
 - Nutritional deficiencies
 - Mental disorders
 - Maternal and child health
 - Congenital anomalies
 - Cancer
 - Occupational health
 - Environmental health
 - Health behaviors

All disease conditions and
related events

Scope...

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Genetic markers of disease risks



Health related behavior



Chronic diseases,
injuries,
birth defects

MCH

Occupational health

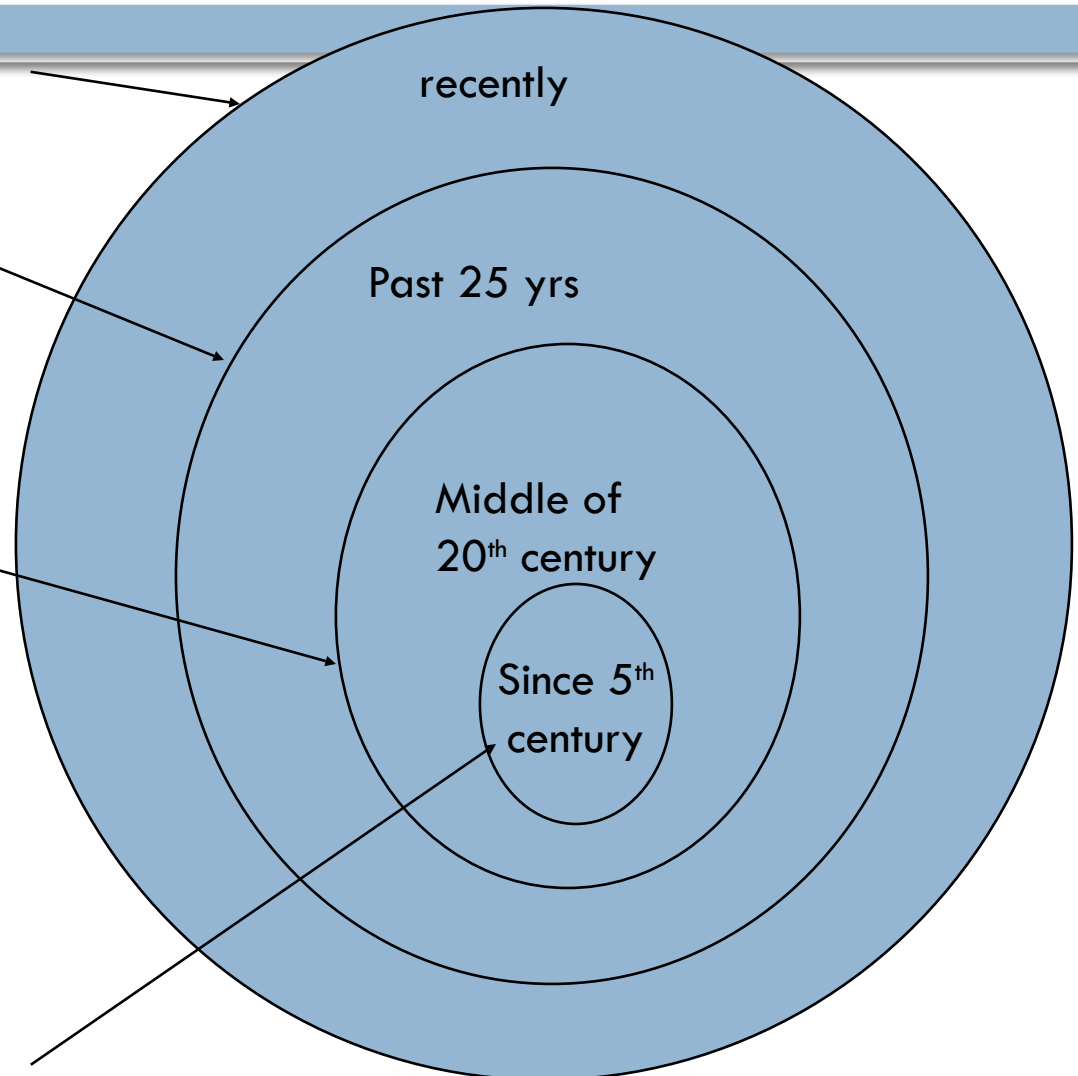
Env'tal health



Endemic communicable diseases &
Non communicable diseases



Epidemic of communicable diseases



Fundamental Assumptions In Epidemiology

60

A. “Disease occurrence in human population is **not random**”

- Disease is not randomly distributed throughout a population

- Epidemiology uses systematic approach to study the differences in disease distribution in subgroup

- Why certain individuals/group acquire disease and others not?

B. “Human disease has **causal and preventive** factors

- That can be identified through systematic investigation of different population or sub group of individuals within a population at different place or time”.

- Allows for study of causal and preventive factors.

◇ **Essential assumption of epidemiology and types of bias in assumption (conceptual error).**

Theories & models of diseases causation

61

- Different concepts on causation of diseases(Ancient and Modern) views, ancient views includes *Demonic, Punitive, Humoral*(Supernatural), *Miasmatic, & Contagion theory*
- Single germ theory
- Epidemiological triad(AHE)
- Epidemiological tetrad(AHET)
- BEINGS theory(complex interplay of nine different factors)
- Web of causation theory
- Wheel theory
- Multifactorial causation(Rothman pie model)

Disease causation theory/model...

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Disease causation

- ❖ The cause of a disease is an event, condition or characteristics that precede the diseases event and without which the disease event either would not have occurred at all or would not have occurred some latter time.
 - Not all associations between exposure and diseases are causal.
 - If disease does not develop without the factor being present, then we term the causative factor “**necessary**” if the disease always result from the factor then we term the causative factor “**sufficient**”
 - Example- Tubercle bacilli is a necessary factor for tuberculosis
 - Rabies virus is sufficient for developing clinical rabies.

Theories...

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Principles of causation

- There are two principles of disease causation
 - The single germ theory
 - The ecological approach

Germ theory

- **First proposed by Girolamo Fracastro (1546).**
- **Luis Pasteur (1822-1895)** isolated microorganisms.
- This discovery led to R. **Koch's postulate** in 1877.
- Koch's postulate states that
 - The organism must be present in every case
 - The organism must be isolated and grown in culture
 - The organism must, when inoculated in to a susceptible animal, cause the specific disease.
 - The organism must then be recovered from the animal.

The Single Germ Theory_Henle-koch Postulates

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Sometimes called “pure determinism”

- Each disease will be caused by a germ
- Without that germ that disease will not be caused
- By introducing that germ , that disease can be caused in animals experimentally,
- And that germ can again be isolated from that sick animal experimented with.

Theories...

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Ecological approach

- Ecology is the study of the relationship of organism to each other as well as to all other aspects of the environment
- The requirement that more than one factor be present for disease to develop is referred to as **multiple causation or multi-factorial etiology**.
 - In the ecological view, an agent is considered to be necessary but not sufficient cause of a disease because the conditions of the host and environment must also be optimal for a disease to develop.
 - Example- Mycobacterium tubercle bacillus is a necessary but not sufficient cause for tuberculosis.
- Etiology of a disease includes all factors that contribute to the occurrence of a disease.
 - These factors are related to **agent, host, and environment**.

General Models of Causation

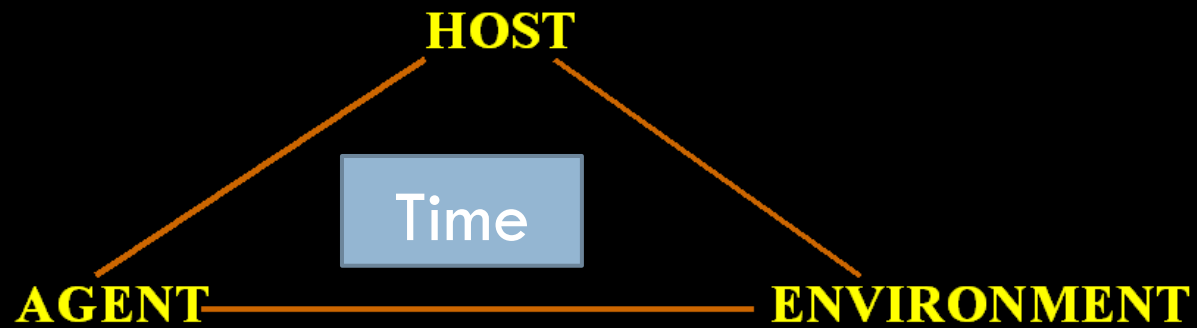
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- ❖ In epidemiology, there are several models of disease causation that help to understand disease process.
- ❖ The most widely applied models are:
 - The epidemiological triad (*triangle*)
 - The *wheel*
 - The *web*.
 - The sufficient cause and component causes models (Rothman's component causes model)

The epidemiological triad and Tetrad

67

The Epidemiologic Triad



Agent factors

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- ❑ **Infectious agents:** agent might be microorganism—virus, bacterium, parasite, or other microbes. e.g. polio, measles, malaria, tuberculosis
 - ❑ Generally, these agents must be present for disease to occur.
- ❑ Nutritive: excesses or deficiencies (Cholesterol, vitamins, proteins)
- ❑ Chemical agents: (carbon monoxide, drugs, medications)
- ❑ Physical agents (ionizing radiation,...

Host factors

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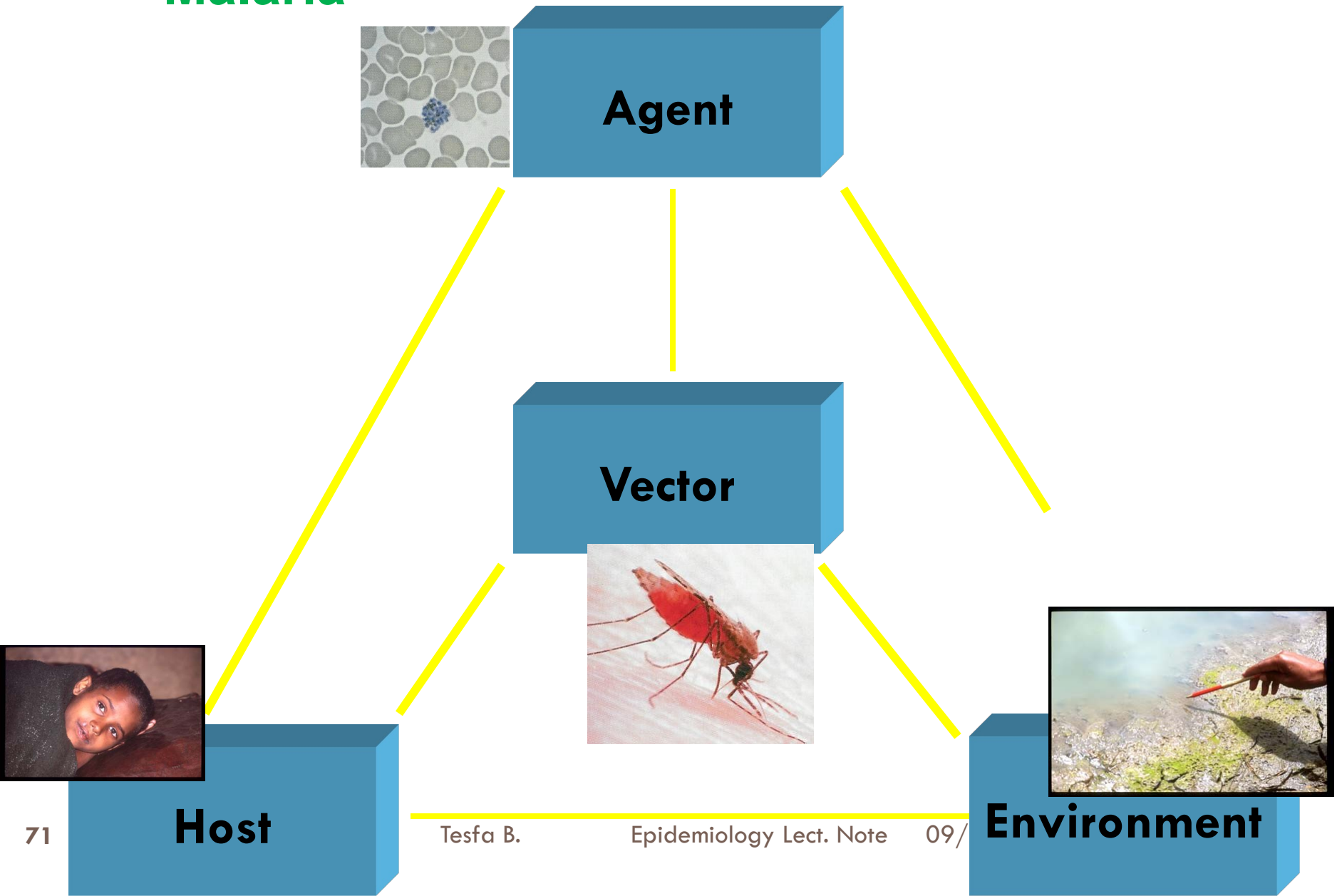
- Host factors are intrinsic factors that influence an individual's **exposure, susceptibility, or response** to a causative agent.
- Host factors that affect a person's risk of exposure to an agent:
 - e.g. Age, race, sex, socioeconomic status, and behaviors (smoking, drug abuse, lifestyle, sexual practices and eating habits)
- Host factors which affect susceptibility & response to an agent:
 - Age, genetic composition, nutritional and immunologic status, anatomic structure, presence of disease or medications, and psychological makeup.

Environmental factors

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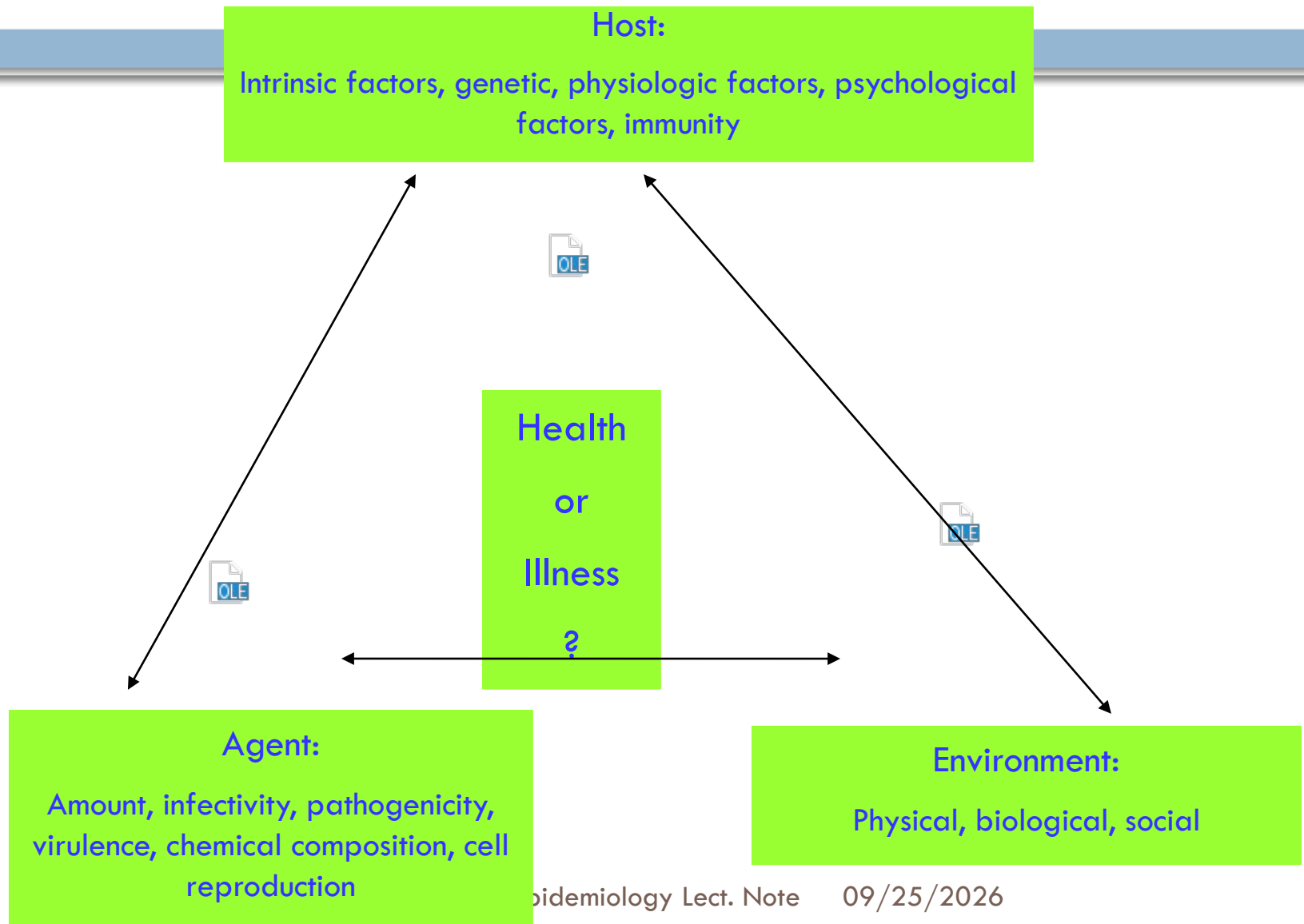
- Environmental factors are extrinsic factors which affect the agent and the opportunity for exposure
- Environmental factors include:
 - ▣ Physical factors such as geology, climate,..
 - ▣ Biologic factors such as insects that transmit an agent; and
 - ▣ Socioeconomic factors such as crowding, sanitation, and the availability of health services.

Malaria



The epidemiologic triad Model

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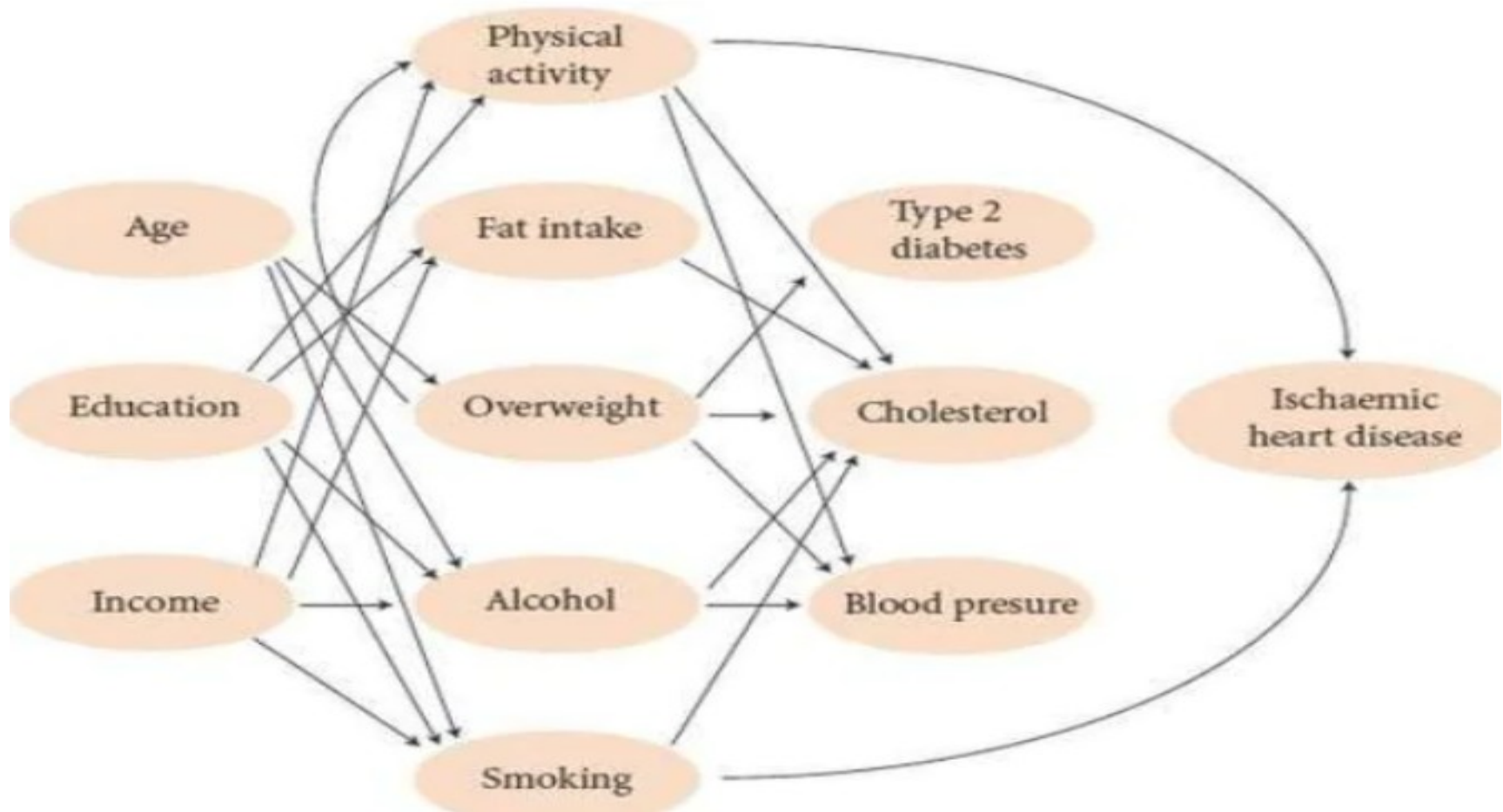
Web of Causation

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- ❖ Given by **Brian MC Mohan and Pugh**
- ❖ Given for CDs where the **cause is unknown** but is the outcome of the **interaction** of multiple factors.
- ❖ All predisposing factors along with inter-r/ships.
- ❖ There is no single cause
- ❖ Causes of disease are interacting
- ❖ Illustrates the interconnectedness of possible causes
 - Developed to de-emphasis agent
 - Chain of causation
 - Complexity of origin is web
 - Multiple factors promote or inhibit
- ❖ Emphasizes multiple **interactions** between host and environment.

Web of causation...

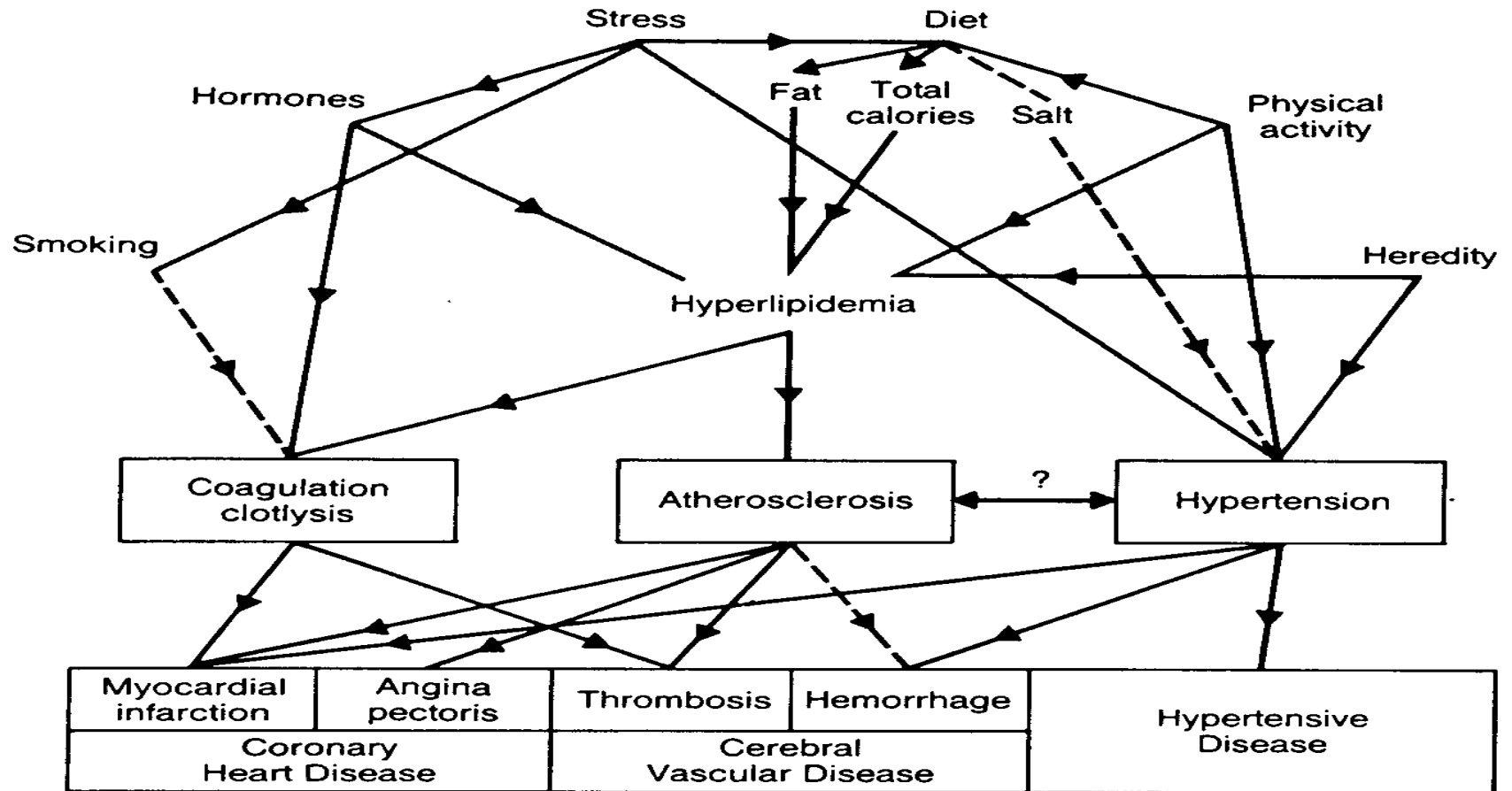
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Web of causation...

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Web of Causation for the Major Cardiovascular Diseases



The Wheel of Causation

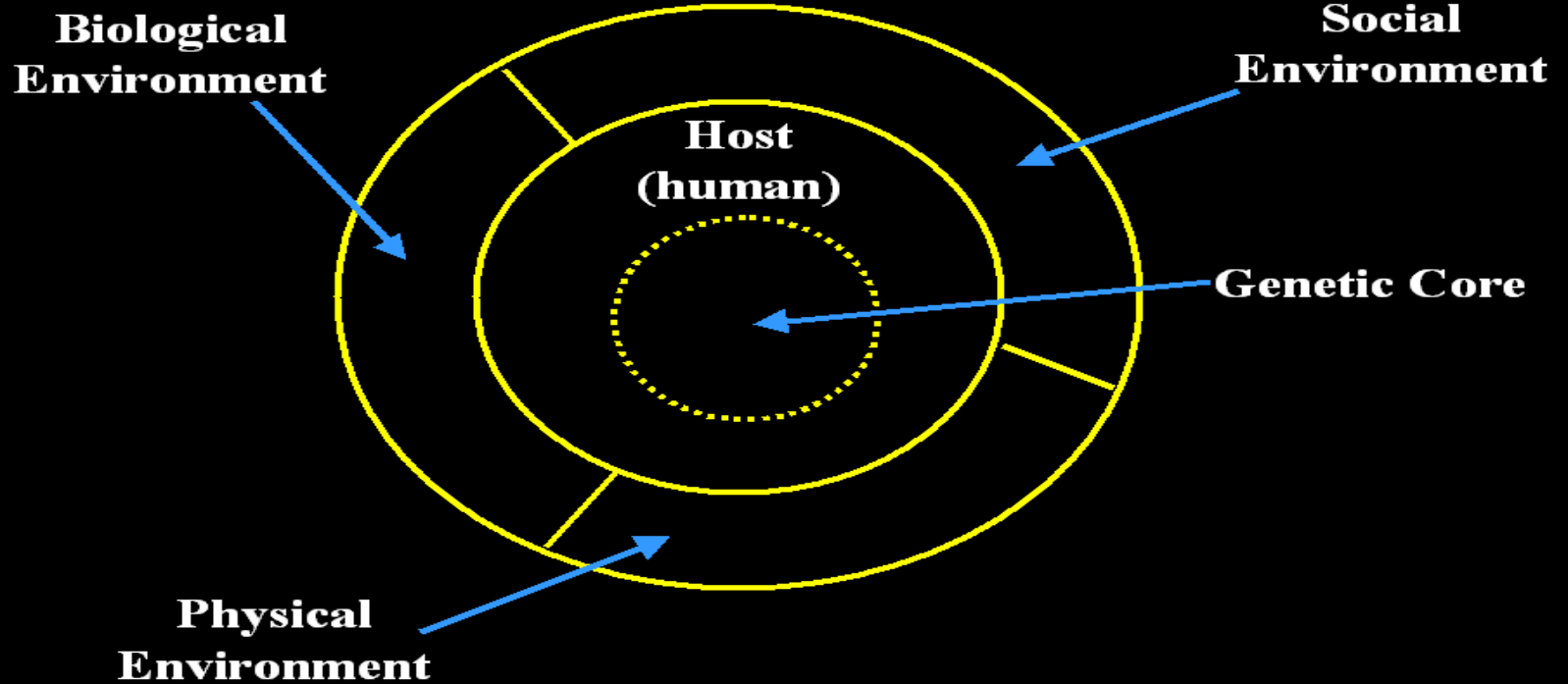
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- ❖ The Wheel of Causation de-emphasizes the **agent** as the **sole** cause of disease
- ❖ It emphasizes the interplay of **physical, biological and social** environments.
- ❖ It also brings **genetics** into the mix.

The Wheel of Causation

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The Wheel of Causation



Multifactorial causation theory

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- Dr. **Max Joseph von Pettenkofer**
(**German**-Bavarian chemist & **hygienist**)
- Contradicted the unifactorial theory of disease causation and emphasized that disease is Multifactorial in causation.
- Several causes or factors acting jointly, cumulatively, by complementing or in an unexplained manner will lead to the disease.
- He is known for his work in practical hygiene, as an apostle of good water, fresh air and proper sewage disposal



Causal Relationships

79

A causal pathway may be direct or indirect:

- Direct causation, A causes B without intermediate effects
- Indirect causation, A causes B, but with intermediate effects
- In human biology, intermediate steps are virtually always present in any causal process

Types of causes

80

❖ Sufficient causes:

- ▣ **Sufficient** cause is a “minimum set of conditions, factors or events needed to produce a given outcome.
- ▣ A set of conditions without any one of which the disease would not have occurred
- ▣ Not usually a single factor, often several

❖ Necessary cause:

- ▣ A **necessary** cause is a causal factor whose presence is required for the occurrence of the effect. If disease does not develop without the factor being present, then we term the causative factor “**necessary**”.
- ▣ Must be present for disease to occur, disease never develops in the absence of that factor.
- ▣ A **component cause** that is a member of every sufficient cause.

Types of causes...

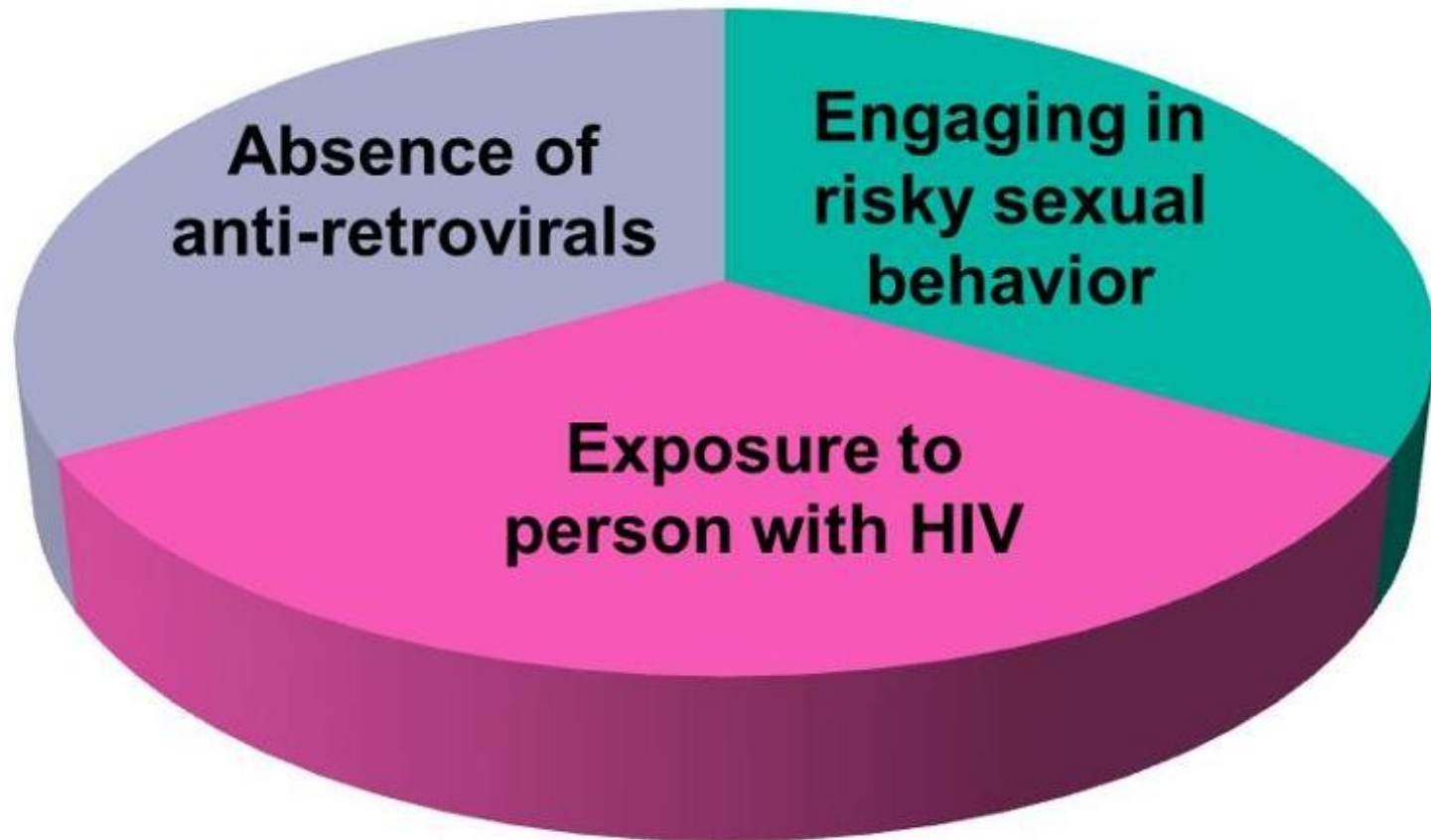
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Types of Causal Relationships

- **Necessary and sufficient** – without the factor, disease never develops
 - With the factor, disease always develops (this situation rarely occurs)
- **Necessary but not sufficient** – the factor in and of itself is not enough to cause disease
 - Multiple factors are required, usually in a specific temporal sequence (such as carcinogenesis)
- **Sufficient but not necessary** – the factor alone can cause disease, but so can other factors in its absence
 - Benzene or radiation can cause leukemia without the presence of the other
- **Neither sufficient nor necessary** – the factor cannot cause disease on its own, nor is it the only factor that can cause that disease
 - This is the probable model for chronic disease relationships

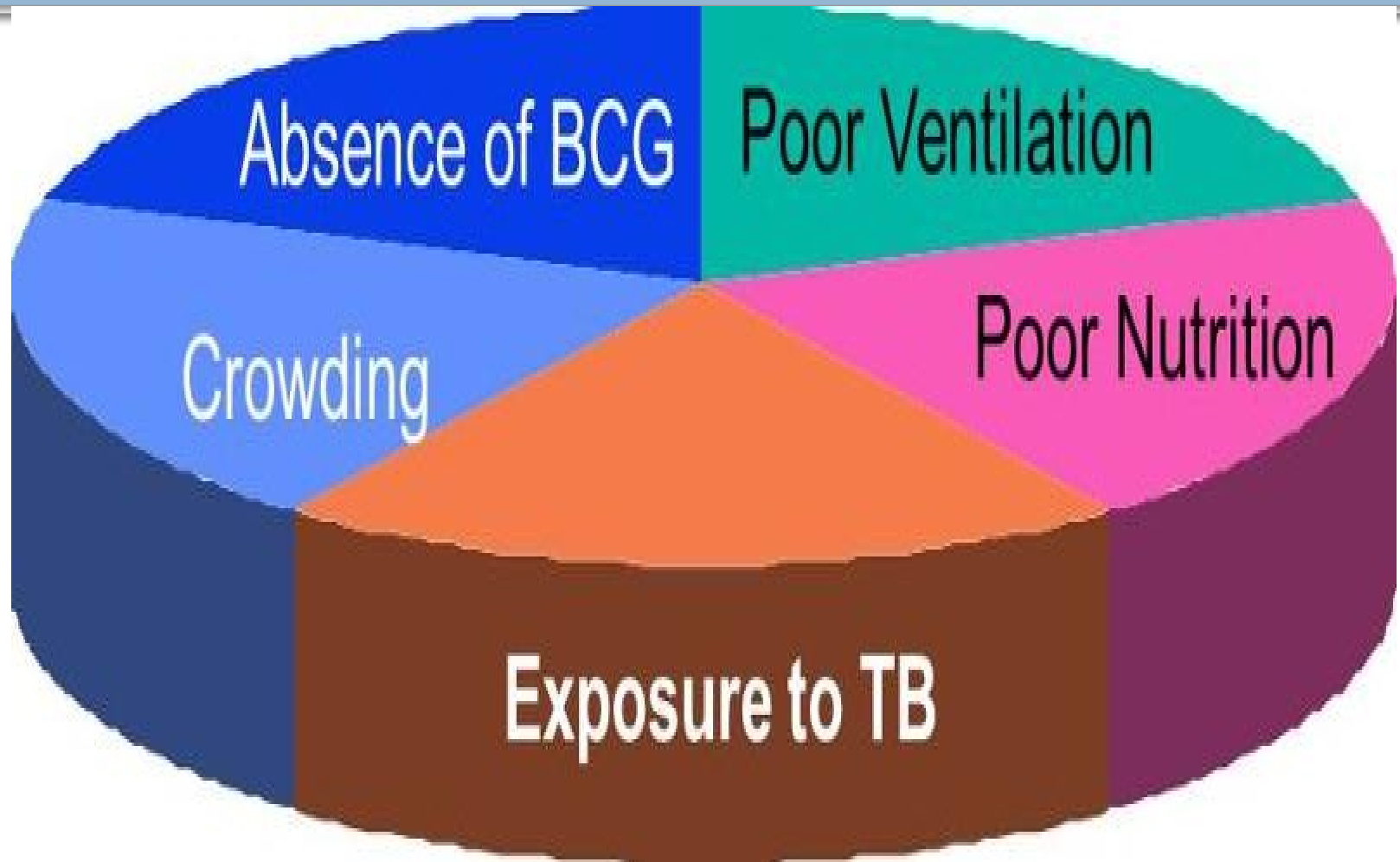
The sufficient cause and component causes model **Rothman's component causes model**

82



Sufficient-component cause model to tuberculosis (TB), one possible cause might be represented by the pie chart below.

83



Sufficient-component cause model to tuberculosis (TB), one possible cause might be represented by the pie chart below.

- Among the three sufficient causes of TB illustrated above, there are both similarities and differences in the composition of the components.
- They also differ in the number of components.
- For example, an individual with AIDS and poor nutrition would be severely immunocompromised, so the only component needed to complete the causation of TB would be exposure to the TB bacillus.

Features of the Sufficient-Component Cause Model

85

Aschengrau and Seage point out some of the key features of the sufficient-component cause model:

- A cause is not a single component, but a minimal set of conditions or events that inevitably produces the outcome.
- Each component in a sufficient cause is called a **component cause**.
- There may be a number of sufficient causes for a given disease or outcome.

Features of the Sufficient-Component Cause Model...

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- A component cause that must be present in every sufficient cause of a given outcome is referred to as a necessary cause. For example, HIV exposure is necessary for AIDS to occur, and TB exposure is necessary for TB infection to occur.
- The completion of a sufficient cause is synonymous with the biologic occurrence of the outcome, e.g., the transition to a malignant cancer within a single cell marks the biologic onset of the cancer.
- The components of a sufficient cause do not need to act simultaneously; they can act at different times.

Rothman's model...

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- The factors or conditions that form a sufficient cause are called **component** causes.
- Example
 - ▣ The tubercle bacillus is required to cause tuberculosis but, alone, does not always cause it,
 - ▣ so tubercle bacillus is a **necessary**, not a sufficient, cause.
- Rothman's model has emphasised that the causes of disease comprise a collection of factors.
 - ▣ These factors represent pieces of a pie, the whole pie (combinations of factors) are the **sufficient** causes for a disease.
 - ▣ It shows that a disease may have more than one sufficient cause, with each sufficient cause being composed of several factors.

Rothman's model...

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- ▣ The factors represented by the pieces of the pie in this model are called **component** causes.
- ▣ Each single component cause is rarely a **sufficient** cause by itself, But may be **necessary** cause.
- ▣ Control of the disease could be achieved by removing one of the components in each "pie" and if there were a factor common to all "pies" (**necessary cause**) the disease would be eliminated by removing that alone.
- ◇ **Disease causation theory and models**

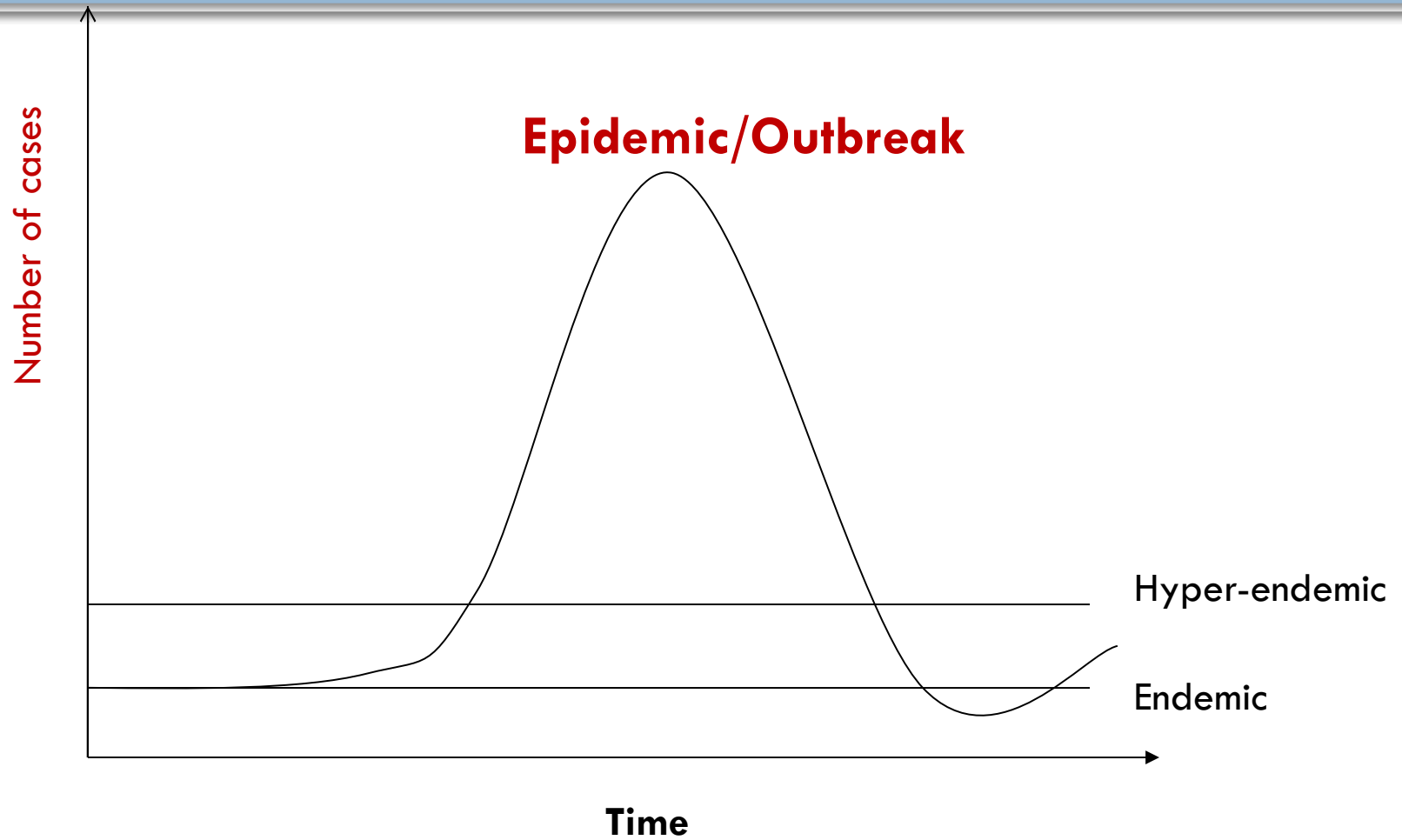
Levels of disease occurrence

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- ❖ Diseases occur in the community at different levels at a particular point in time.
- ❖ Some diseases are usually present in a community at a certain predictable level; this is called the **expected level**.
- ❖ But at times, diseases may occur in **excess of what is expected**.
 - ▣ **Expected level**
 - **Endemic**-a persistent level of low to moderate occurrence of diseases
 - **Hyper endemic**-a persistently high level of occurrence of disease
 - Sporadic-occasional cases occurring at irregular intervals
 - ▣ **Excess of what is expected**
 - **Epidemic-occurrence** of diseases in excess of what is expected in a limited period
 - **Outbreak**-same as epidemic often used by public health officials because it is less provocative to the public
 - **Pandemic**-an epidemic spread over several countries or continents, affecting a large number of people.

Levels...

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Verities of Epidemiology

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□ As Epidemiology continues to develop and to expand into new areas, the field has diversified into many forms:

- **Shoe-leather** epidemiology-The term is often synonymous with *field* or *intervention epidemiology*.
 - All 3 terms imply investigations initiated in response to urgent public health problems and for which the investigative team does much of its work outside the office or laboratory).
- **Microbial Epidemiology** – biology and ecology of pathogenic microorganisms, their lifecycles, and their interactions with their human and non-human hosts
- **Descriptive epidemiology** – examination of patterns of occurrence of disease and injury and their determinants
- **“Risk factor” Epidemiology** – searching for exposure-disease associations that may provide insights into etiology and avenues for prevention
- **Clinical Epidemiology**– the application of epidemiological principles and methods to the practice of CM/ it assess accuracy, efficacy, effectiveness, and unintended consequences of methods of prevention, early detection, diagnosis, treatment, and management of health conditions.

Verities....

92

- **Molecular Epidemiology** – investigate disease at the molecular level to precisely characterize pathological processes and exposures, to elucidate mechanisms of pathogenesis, and to identify precursor conditions
- **Genetic Epidemiology** – the confluence of **molecular biology**, **population studies**, and **statistical models** with an emphasis on heritable influences on disease susceptibility and expression
- **Big Epidemiology**** – multisite collaborative trials, such as the Hypertension Detection and Follow-up Program (HDFP), Coronary Primary Prevention Trial (CPPT), Multiple Risk Factor Intervention Trial (MRFIT), Women's Health Initiative (WHI)
- **Entrepreneurial Epidemiology** – building institutions and careers by winning research funding and facilities.

Verities...

93

- **Testimonial Epidemiology** – giving depositions and testifying in court or in legislative hearings on the state of epidemiologic evidence on a matter of dispute
- **Nutritional Epidemiology:** is a special area or discipline of epidemiology that deals with nutritional determinants of diseases.
- **Social Epidemiology** – interpersonal and community-level factors influencing health at the population level.

Verities...

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- ❑ **Reproductive Epidemiology:** Is the study of distribution and determinants of health related states or event in a human populations an the application of this study to promote complete physical, mental, social well-being as it related to reproductive system and to its functions and processes
- ❑ **Global Epidemiology** – assessing the effects of human activity on the ecosystem that supports life on Earth.
- ❑ **Environmental Epidemiology:** A subspecialty of epidemiology that uses epidemiological principles, reasoning, and methods to study the health effects on populations of exposure to all that which is external to the individual human host (physical, chemical, and biological agents) and of immediate and remote social, economic, and cultural factors.

Verities...

95

- **Infectious diseases epidemiology:** The study of circumstances/condition under which both infection and disease occur in a population and the factors which influence their frequency, spread and distribution of infectious diseases
- **Occupational Epidemiology:** The study of the effects of workplace exposures on the frequency and distribution of diseases and injuries in the population. The application of epidemiological knowledge to populations of workers.

Natural history of disease and levels of prevention

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Contents

- Natural history of disease
- Stages in the natural history disease
- Levels of disease prevention
- Applications to common diseases

Natural history of diseases

97

- It is the course of a disease from pathological **onset or inception** to **resolution/termination** over time, **in the absence of intervention/Rx/Prevention**.
 - ▣ Are the health outcomes in ill persons who are not receiving a therapy that influence the rate of the outcomes.
 - ▣ **It is the course of illness in the absence of intervention**
- Is the major elements of **descriptive** epidemiology.
- Best established by **cohort** study.
- Since it is costly and laborious, largely based on other epidemiological studies such as cross sectional and retrospective designs.

Stages in the Natural History of Diseases

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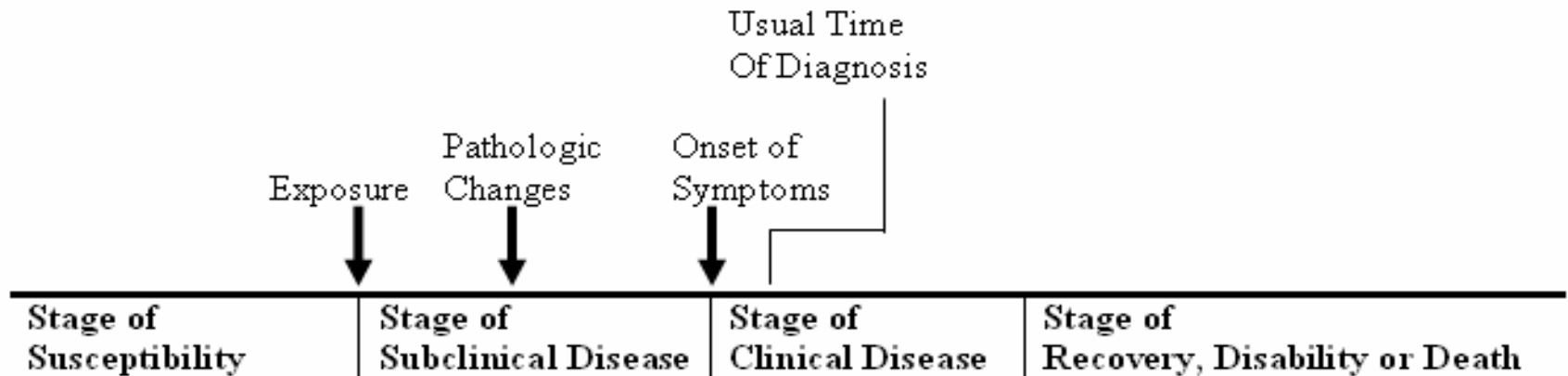


Figure. Natural History of Disease

Individual appraisal#

- ❖ At what level of disease prevention screening found, and best suited level?
- ❖ What is early detection of disease?
- ❖ Does detection before the usual clinical diagnosis guarantee diagnosis at an early stage, ,does it assure a more effective treatment?

Natural History of Diseases (#4 stages)

99

1. **Stage of susceptibility** - this is a stage in which disease has not developed but the ground work has been laid by the presence of factors that favor its occurrence.
 - ▣ Susceptible person is lacking sufficient resistance to particular pathogenic agent to prevent disease if or when exposed.
 - ▣ Under this stage there might be the following
 - ▣ **Incubation period**- is the time interval between entry and development of signs and symptoms of disease.
 - It is the time interval between the entry of pathogenic micro organism and sufficient multiplication to cause pathology that will produce signs and symptoms of disease.
 - ▣ IP in infectious diseases, is equivalent to combination of induction and latent periods in non infectious diseases.
 - **Induction period** is the period of time from causal action until disease initiation.
 - **Latent period**- is the period from asymptomatic disease to development of symptoms. E.g: Hep (2-6weeks), leukemia due to bomb blast 2-12 peaked 6-7 years.

Stages...

100

2. Stage of pre-symptomatic disease or sub-clinical stage

- ▣ There is **no** manifested disease
- ▣ In other words no S/S are detected
- ▣ The person does not know that he has any disease
- ▣ The sub-clinical stage of disease may lead to the clinical stage or the individual may recover.
- ▣ The probability of developing symptoms or disease after becoming infected is the **pathogeneity** of the interaction of the parasite with the host
- ▣ **Pathogeneity** is the ratio of clinical cases to sub-clinical cases.
- ▣ An inapparent infection or silent infection is a successful infection that does not develop detected symptoms
 - Unapparent cases can be infectious.

Stages...

101

3. The clinical stage

- ▣ In this stage the person has s/s of disease
- ▣ There are various grades of illness with different outcomes depending on the agent host interaction
- ▣ Some diseases are short and mild that every one recovers quickly while others are very serious leading to complications and death
- ▣ Examples:
 - ▣ Rabies has a *relatively short but severe clinical stage* and almost always results in ***death***
 - ▣ Diabetes Mellitus has a *relatively longer clinical stage* and eventually results in ***death if the patient is not properly*** treated.

Stages...

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4. Stage of disability

- Some diseases run their course and then resolve completely either **spontaneously** or under the influence of **therapy**.
- However, there are a number of conditions which give rise to residual defects of **short** or **long** term duration leaving the person disabled to a greater or lesser extent.

Levels of Prevention

103

- The important tasks of health workers are preventing people from getting sick, from getting worse, becoming disabled or becoming chronically sick or dying.
- Prevention is an actions aimed at eradicating, eliminating, or minimizing the impact of disease and disability, or if none of these is feasible, retarding the progress of disease and disability.
- There are **four** important levels of prevention:
- Primary , secondary and tertiary [**Traditional** classification].
- Other levels (primordial and quaternary prevention)
- There is significant conceptual and practical overlapping among levels largely, depending on the type of disease

1. Primordial prevention

- Existence of underlying condition leading to causation
- The aim is to avoid the emergence and establishment of the social, economic, and cultural patterns of living that are known to contribute to an **elevated risk of disease**.
- **Target:** Total population and selected groups
 - Example: smoking, environmental pollution.

Level...

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2. Primary prevention- is preventing healthy people from becoming sick.

- In order to carry out effective primary prevention, we must ask **why healthy people become sick**.
- We must know who most at risk of getting the disease.
 - ▣ It includes health promotion, specific protection and prevention of disease occurrence.
- Some of primary prevention are:
 - ▣ Immunization
 - ▣ Provision of safe water supply
 - ▣ Health related behavioral change through health education
 - ▣ Provision of adequate housing
 - ▣ Safe disposal of human excreta
 - ▣ Control of vectors of diseases
 - ▣ Protection against accidents and occupational hazards
 - ▣ Good nutrition, adequate clothing, rest and recreation.

Level...

105

3. Secondary prevention

- It involves detecting people who already have the disease as early as possible in order to cure the patient and prevent chronic disease and disability [**early clinical detection of cases detection**].
- When primary prevention has failed, a healthy person becomes sick.



Level...

106

- Secondary prevention thus involves the **detection and treatment of those people in the sub clinical stage**
- Early detection depends on the **clinical stage and availability of investigation facilities**
- **Limit severity and spread of disease, that is decrease secondary cases**

4. Tertiary prevention

- Is the prevention of more disability and death
- The aims are:
 - Disability limitation and Rehabilitation
 - In a patient who can not be cured, tertiary prevention have many objectives:

Level...

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Objectives of rehabilitation:

- ▣ **Restore function-** e.g. surgical correction of leprosy => medical rehabilitation
- ▣ **Vocational training-** so that the individual can earn his/her livelihood=> vocational rehabilitation
- ▣ **Integrate family with society** – social rehabilitation
- ▣ **Instill self confidence-** psychological rehabilitation

5. Quaternary prevention

- ▣ Concept coined by the Belgian general practitioner Marc Jamouille, are the actions taken to identify a patient at risk of overmedicalisation, to protect them from new medical invasion, and to suggest interventions which are ethically acceptable
- ▣ Quaternary prevention consists of actions that identify patients at risk of over diagnosis or overmedication and that protect them from excessive medical intervention.
- ◇ Actions that prevent iatrogenesis.

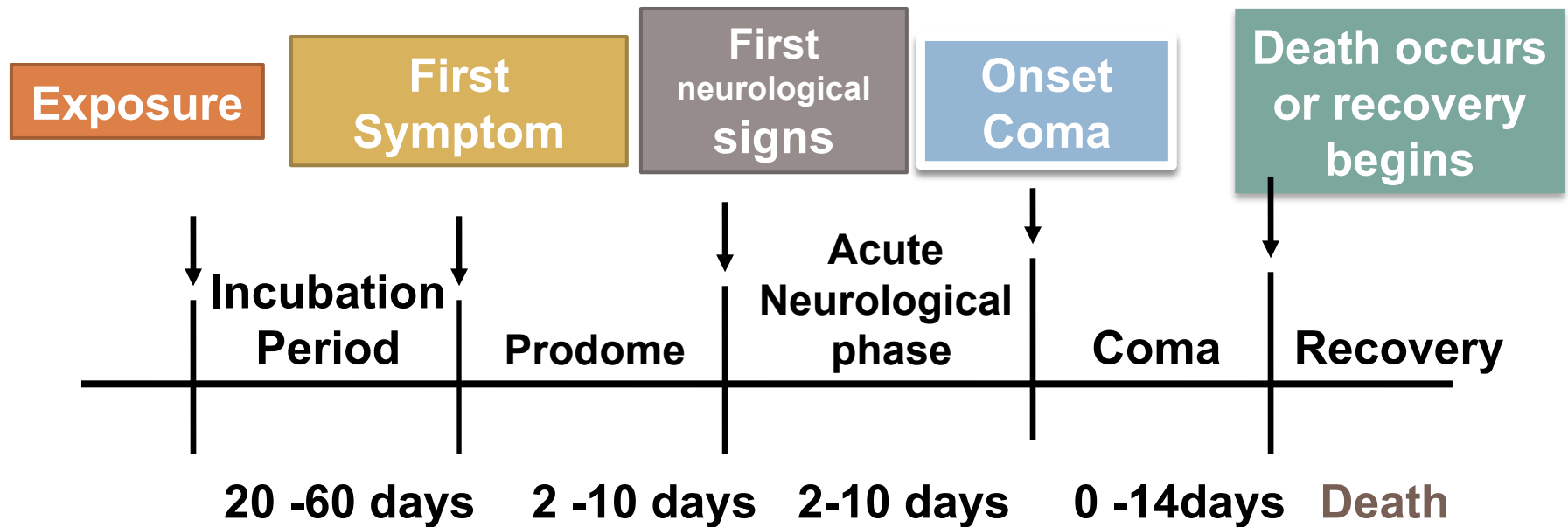
Determinants of Prevention

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- Successful prevention counts on:
 - ▣ A knowledge of causation,
 - ▣ Dynamics of transmission,
 - ▣ Identification of risk factors and risk groups,
 - ▣ Availability of prophylactic or early detection and treatment measures,
 - ▣ An organization for applying these measures to appropriate persons or groups, and
 - ▣ Continuous evaluation and development of procedures applied.
- ▣ **Is spectrum of disease resemble to NHxDs?**

Rabies in Humans

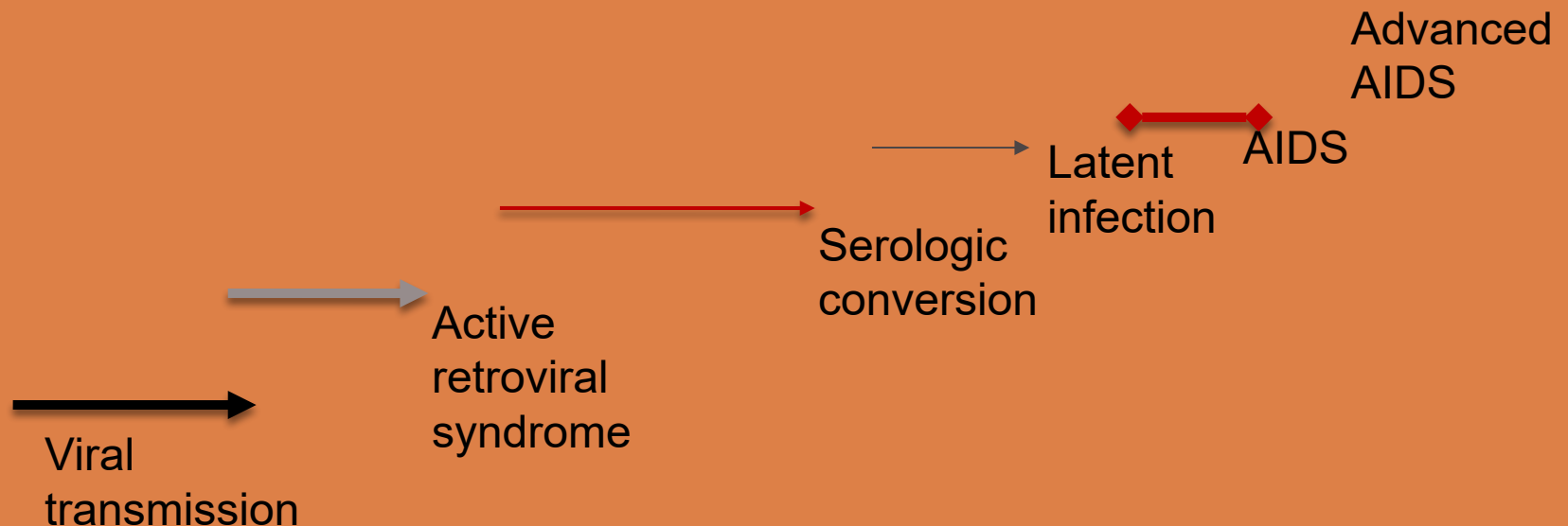
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The natural History of HIV infection, untreated infection with HIV causes a spectrum of clinical problems beginning at the time of seroconversion (primary HIV) and terminating with AIDS and usually death.

110

Has six stages



Levels of disease

111

Level of prevention	Stage of disease	Target
1. primordial	Existence of underlying condition leading to causation. The aim is to avoid the emergence and establishment of the social, economic and cultural patterns of living that are known to contribute to an elevated risk of disease. Eg. Smoking, environmental pollution	Total population and/or selected groups
2. primary	Specific causal factors exist. The causative agent exists but the aim is to prevent the development of disease. Example: immunization, measles, polio	Total population selected groups and healthy individuals
3. secondary	Early stage of disease. The aim is to cure patients and prevent the development of advanced disease Eg. Early detection and treatment of cases of TB and STI	patients
4. Tertiary	Late stage of disease (treatment and rehabilitation). The aim is to prevent sever disability and death. Example: leprosy	Patients

2. The Infectious disease cycle

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Contents

- Chain of disease transmission
 - ▣ Time course of an infectious diseases
 - ▣ **Carrier** in infectious disease transmission
 - ▣ **Estimating transmission probability**
 - ▣ Secondary attack rate
 - ▣ Binomial Model of Transmission probability
 - ▣ Basic Reproductive Number (R_0)
 - ▣ Effective reproductive Number (R)
 - ▣ Vaccination
 - ▣ Herd immunity
 - ▣ Performance indicators of vaccination program
 - ▣ Principles of communicable disease control

Basic concepts

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- **Infection/colonization:** Entry & development/ multiplication of disease causing micro organisms in man or animals.
- May be **contained or progress** to disease depending on the immunity
- **Latent infection:** the organism is not being shed, but may get reactivated after some time e.g. HSV (herpes simplex virus)
- **Pathogeneity:** is the probability of developing symptoms or disease after becoming infected/**host-agent interaction**
- **Nosocomial or hospital acquired infection-** contract infection during his/her stay in hospital e.g. UTI (catheter), AIDS (unsafe injection)

Basic...

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- **Community acquired infection:** infection that occurs in the community/general population (as distinguished from nosocomial infection).
- **Cross-infection-** contracted from fellow patients/hospital environment
- **Iatrogenic diseases (physician induced)-** as a result of treatment or investigation procedures in the hospital e.g. allergy and adverse reaction to drugs
- **Infectious disease-** disease resulting from an infection e.g. leprosy and malaria
 - Thus, this phrase is interchangeably used with the expression communicable disease, as both mean the same.

Basic...

115

- **Infestation**-lodgment, development, and reproduction of arthropods on the surface of the body e.g. pediculosis, scabies
- **Contamination**- presence of infectious agent on body surface, in/on cloths, beddings, toys, surgical instruments, dressings and other inanimate articles like water, milk or food.
- **Communicable disease**- disease caused by specific micro organism which is transmitted from person to person, animal to animal, animal to man and environment to man either directly or indirectly e.g. cholera, typhoid, STD etc.
- **Disinfection**: The procedure of killing infectious agents outside the body by direct exposure to chemical or physical agents.

Basic...

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- ❑ **Disinfestations:** The procedure of destroying or removing undesired small animal forms, particularly arthropods rodents, present upon the person, the clothing, or in the environment of an individual, or on domestic animals using chemical or physical agents.
- ❑ **Pollution-** presence of offensive but not necessarily infectious agent in the environment e.g. water, soil pollution
- ❑ **Contagious-** disease transmitted through physical contact e.g. scabies, STI and leprosy.
- ❑ **Generation Time:** The interval between receipt of infection by the host and the latter's maximal infectivity.
- ❑ This applies to both clinical cases and unapparent infections. With person-to-person transmission of infection, the interval between cases is determined by the generation time.
- ❑ **Infectibility:** The host characteristic or state in which the host is capable of being infected
- ❑ **Infectivity:** The characteristic of the disease agent that embodies capability to enter, survive, and multiply in the host.
- ❑ A measure of infectivity is the **secondary attack rate**.

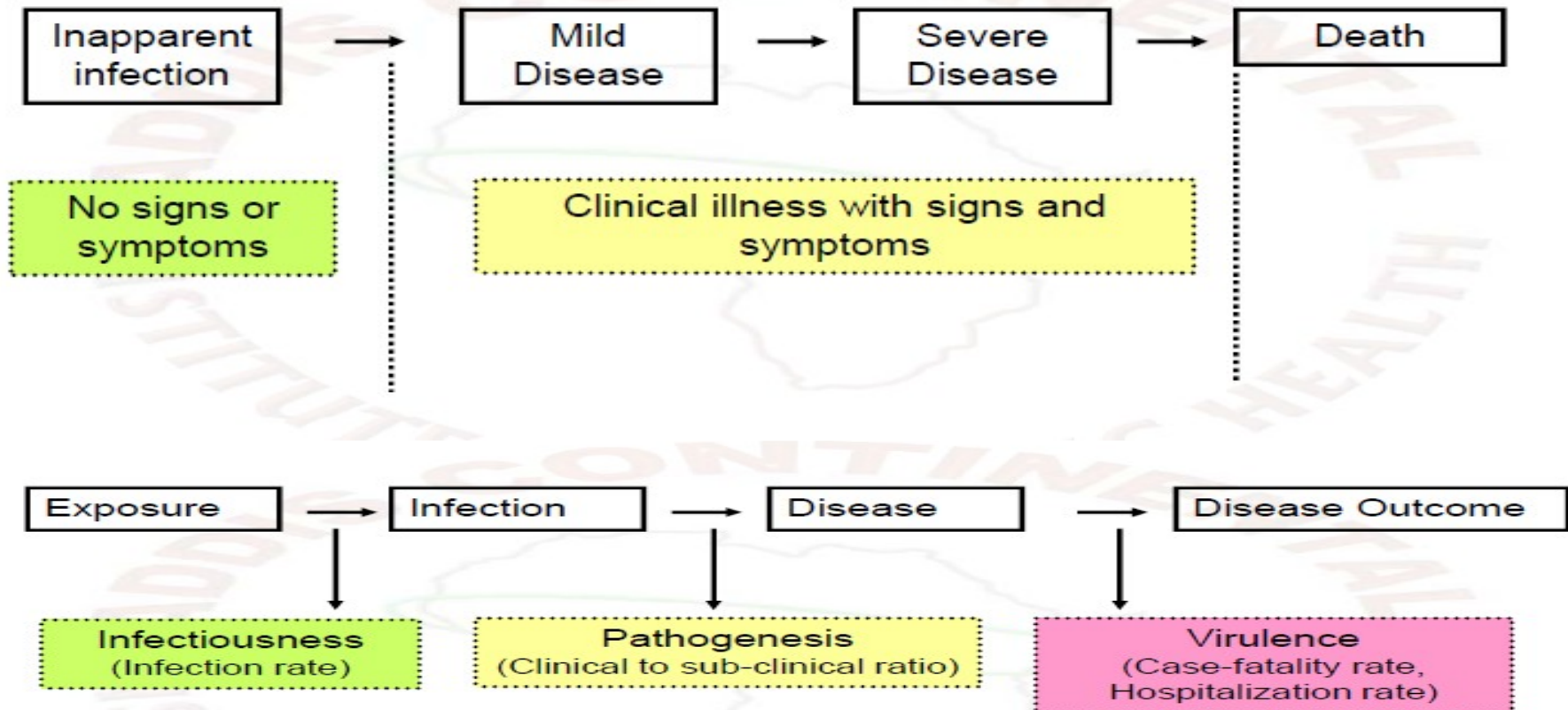
Spectrum of Disease

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- **Spectrum of diseases(SD)**- is the sequence of events that occur in human from the **time of exposure to etiological agent to death**.
- **SD** is a graphical representation of variation in the manifestations of disease.
- It's the full range of manifestations of a disease from precursor states, to subclinical and mild cases, to florid and fulminating disease.
- It's the natural history of a disease from onset to resolution.
- In infectious diseases, this spectrum is known as **gradient of infection**- which reflects the sequence of manifestations of illness in the host reflecting his response to the infectious agent
- **Gradient of infection** is different types of host response to an infection bidirectional extremity.
- **GOI:** The variety of host responses to infection ranging from unapparent infection to fatal illness.
 - **Unapparent or sub-clinical=> presence of an infection in a level below the threshold of clinical symptoms=> mild illness=> moderate illness=> severe/recovery or death**

Spectrum...

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Spectrum...

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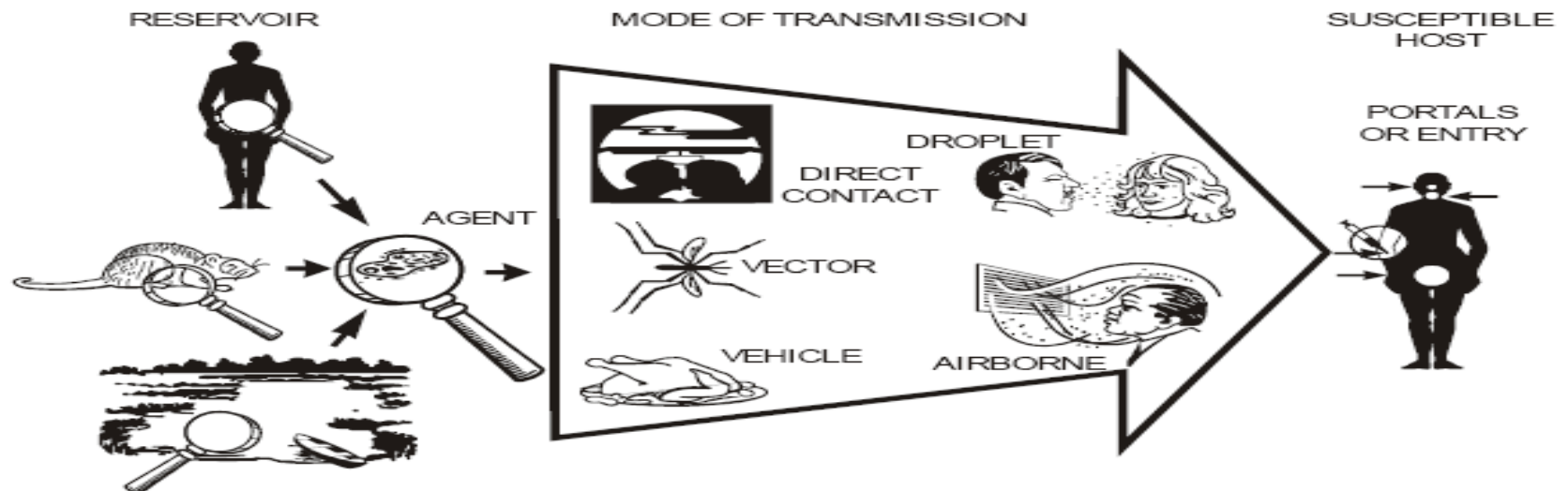
- The frequency with which these different manifestations occur differ with the specific infectious disease- measles 90% apparent, mumps 66% apparent and polio more than 90% in apparent.
 - Hence clinically apparent imply **tip of iceberg** and unapparent imply bottom of iceberg.
 - In fact this situation is called hippopotamus effect or iceberg phenomena.
- **Unapparent infection**- is the presence of an infection below the **threshold** of clinical symptoms.
- The importance are:
 - May exceed clinical cases in number
 - May be more infectious than overt cases
 - Play role in determining the size of the susceptible, since they may induce protective antibody e.g. polio

The Infectious disease Epidemiology

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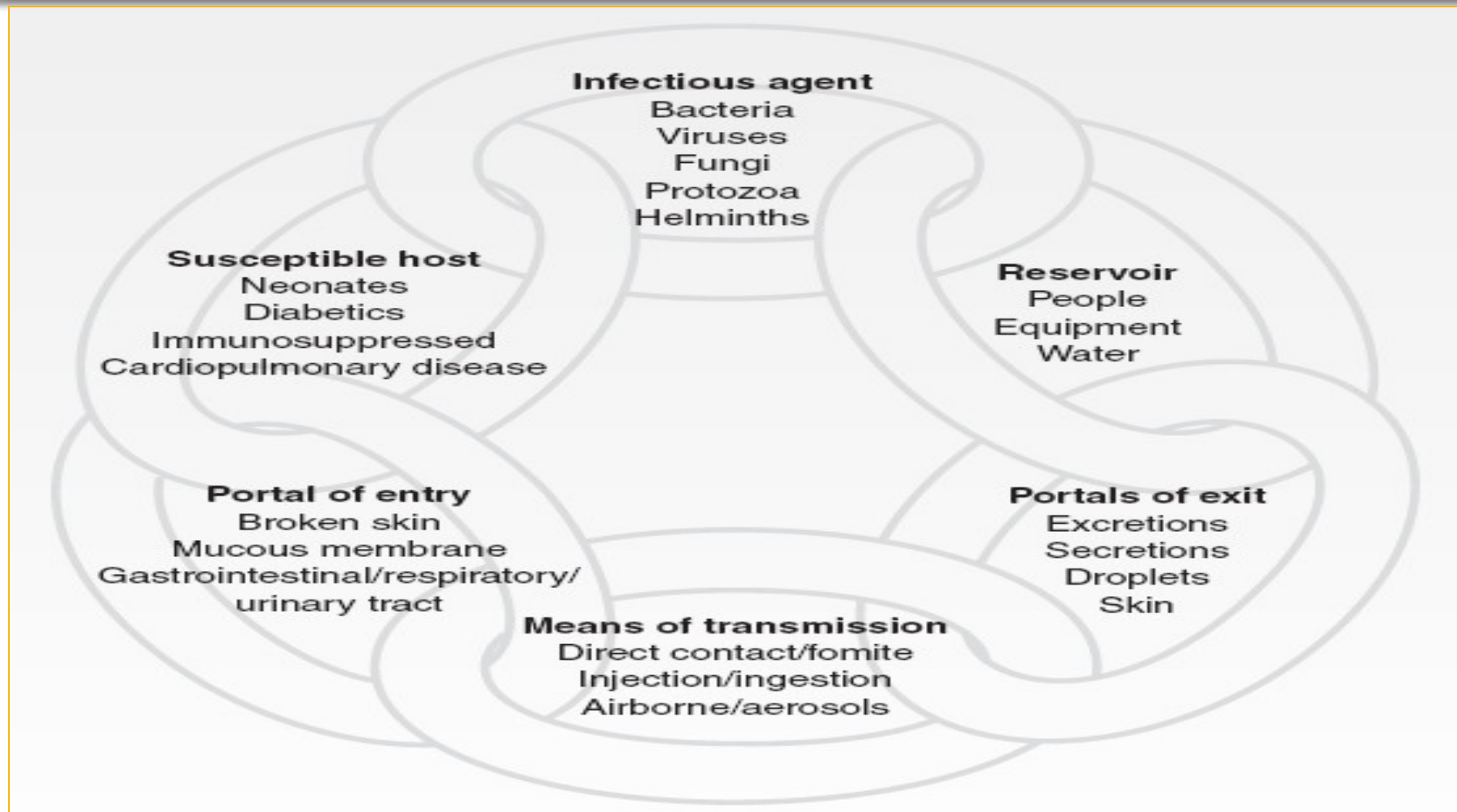
There are **six** components of the infectious disease process constituting chain of disease transmission

1. The agent
2. Its reservoir
3. Its portal of exit
4. Its mode of transmission
5. Its portal of entry
6. Susceptible host



Chain of Disease Transmission

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The Infectious disease cycle...

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- Are chain of factors, that form links in the occurrence and propagation of disease:

1. Infectious agent/etiology

- Is an organism that is capable of producing infection or infectious disease
- The biological properties of the agent may play a major role in its epidemiology
- In order to survive the agent must be able to:
 - **Multiply**
 - **Emerge from the host**
 - **Reach the new host**
 - **Infect the new host**

Major components ...

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- These can generally be grouped into the following five major taxonomic groups
 1. Virus: e.g. HIV; measles; Polio Herpes, etc.
 2. Bacteria: e.g. M. tuberculosis; N. meningitidis; N. gonorrheae
 3. Protozoa: e.g. Plasmodia species; E. histolytica; Giardia lamblia, Leishmania species, etc.
 4. Fungi: e.g. Candida albicans; Cryptococcus neoformans; Histoplasma capsulatum
 5. Metazoa: e.g. Helminths

FACTORS AFFECTING DISEASE occurrence

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- ❑ Microorganism $\neq$ Infectious Agent
- ❑ Exposure $\neq$ Infection
- ❑ Infection $\neq$ Disease
 - ❑ Why exposure to microorganisms is not always followed by infection?
 - ❑ Why infection is not always followed by the development of a manifest disease?
 - ❑ Does infection with a given infectious agent result in a disease of same degree of severity in all individuals?
- ❑ This can be affected by Intrinsic properties of Microorganisms
- ❑ N.B –host agent interaction is characterized by **infectivity, Pathogenicity, virulence or immunogenicity.**

Intrinsic properties of Microorganisms

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Infectivity: is the ability of an agent to enter, invade and multiply (produce infection) in an exposed host

Includes a time range from exposure to infection.

Infectivity is a measure of the progression of an infectious agent from exposure to infection

- It is the ability to cause infection in an exposed susceptible host, and is measured by the infection rate (infectiousness).
- It is the proportion of exposed persons who become infected.

$$\text{Infection Rate} = \frac{\text{Total Number of Infected People}}{\text{Total Number of Suceptible People Exposed}} \times 100\%$$

- A measure of infectivity is the secondary attack rate

Intrinsic...

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- **Infection:** invasion & multiplication in the host,
- **Infectivity:** the proportion of exposed who becomes infected
- $$\text{Infection rate} = \text{Infected} / \text{exposed}$$
- Many **host and environmental factors as well as the dose, the route of entry, source of infection, and the strain of the agent** influence the infectivity of an agent.
- Infectivity and infectiousness don't necessarily imply the same meaning
 - ▣ Infectiousness of a host at a given time in the course of disease is specific for a given disease
 - ▣ Some diseases can be transmitted during the incubation period, some only during the active phase of the disease, and some for years after recovery (chronic carrier state).

Intrinsic...

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- **Pathogenicity:** is the ability of an agent to produce a clinically manifest disease in susceptible host.
 - It is measured by determining the proportion of infections that result in clinically apparent disease
 - Such measurement of pathogenicity is not possible in a population
 - It can, however, be determined by studying a population using laboratory methods during and following an outbreak of a particular disease for which reliable and valid laboratory diagnostic methods are available.



Intrinsic...

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- **Virulence:** this is the ability of an agent to produce severe disease.
 - It is the ability to cause disease in a susceptible host, and can be measured by the clinical to subclinical ratio/the degree of pathogenicity; the disease-evoking power of a microorganism in a given host
 - It is measured by determining the proportion of clinical cases resulting in severe clinical manifestations, including sequelae.
 - When death is the only criterion of severity **case fatality rate (CFR)** **substitute it.**

$$\text{Virulence (CFR)} = \frac{\text{number of fatal cases}}{\text{Total number of cases}} * 100$$

Or

$$= \frac{\text{number of hospitalized cases}}{\text{Total number of cases}} * 100$$

Intrinsic...

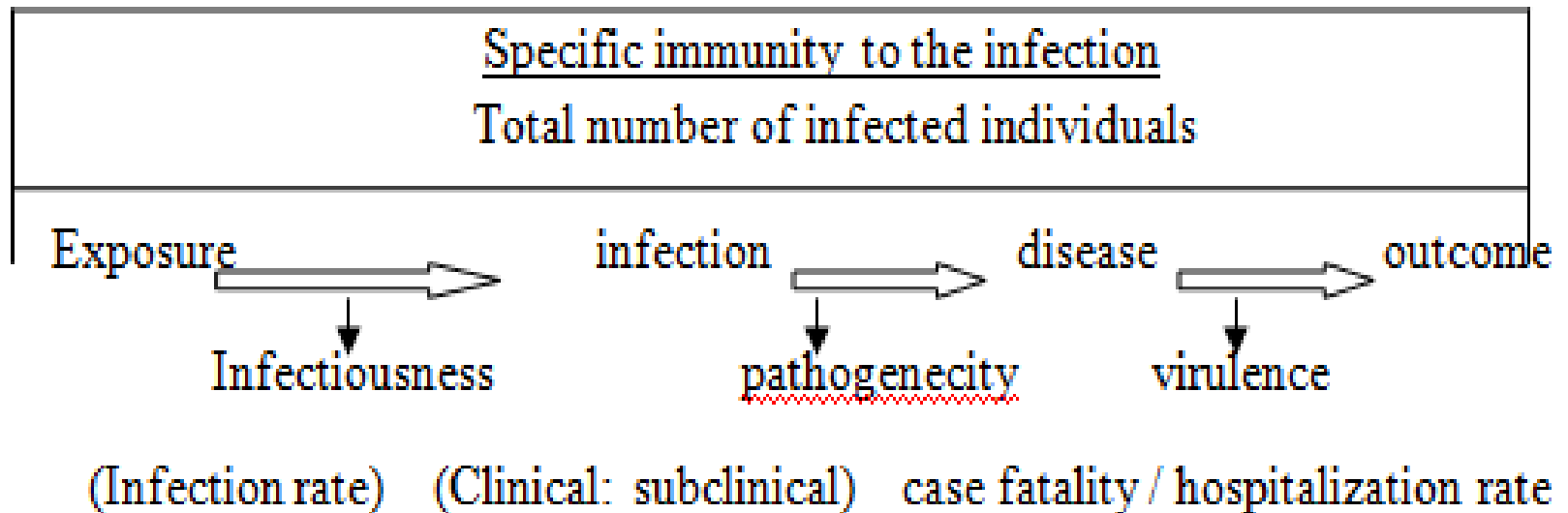
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- High infectivity $\neq$ high pathogenicity;
- High pathogenicity $\neq$ high virulence;
- The definitions used for infectivity, pathogenicity, and virulence as applied to human beings is different from that used in other sciences like microbiology.
 - 📖 Factors that influence the degree of infectivity, pathogenicity, and virulence include:
 - ✍ Strain of agent
 - ✍ Dose of agent
 - ✍ Route of infection
 - ✍ Host factors like host age, host nutritional status; host immune response;
 - ✍ Treatment, especially on virulence; and
 - ✍ Season, through influencing exposure to the agent and other factors that enhance the entry of the agent into the body and its transmission.

Intrinsic...

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- **Immunogenicity:** It is the ability of an infection to produce specific immunity
 - Immunogenicity = $\frac{\text{number of infected individuals who developed}}{\text{Total number of infected individuals}} * 100$.



Component...

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2. Reservoir

- Is the person or animal, arthropod, plant, soil or substance in which an infectious agent can thrive and from where it can be transmitted to another host
- *Types of reservoir*
 - ▣ *Man as a reservoir of infection-hepatitis B virus or Salmonella typhi*
 - ▣ *Animals as a reservoir of infection-brucellosis (cows and pigs), anthrax (sheep), plague (rodents), trichinosis (swine), and rabies (bats, raccoons, dogs, and other mammals).*
 - ▣ *Non-living things as a reservoir of infection*
- *Reservoirs may or may not be similar to source*
- *Reservoirs may be homologous or heterologous*

Component...

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3. **Portal of exit**

- Way in which the agent gets excreted from the host
- This is the site on the reservoir of infection through which the infectious agent escapes from the reservoir. This includes:
 - Gastrointestinal Tract (GIT) e.g. Typhoid fever, amoebiasis etc.,
 - Respiratory Tract e.g. Tuberculosis, common cold
 - Skin and Mucous membrane e.g. STDS, Scabies, ring worm

4. **Mode of spread (Transmission)**

- Means of spread from reservoir to susceptible host
 - This is the mechanism by which an infectious agent is transferred from a reservoir of infection to a new host
- Infectious agent thrive in person, animal, arthropod, plant, soil and substance, and spread to susceptible through source.
 - The material containing an infectious agent through which the actual transfer of the agent from the reservoir to susceptible host takes place e.g. in pulmonary TB, sputum is the source
 - Reservoir: may be person, animal, arthropod, plant, soil or substance.

Component...

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- ▣ Human reservoir may be **overt case, sub clinical case or carrier**
- ▣ Animal reservoir e.g. rabies (dogs), Japanese encephalitis (birds, pigs)
- ▣ Soil: Clostridium tetanus, B. anthracis
- ▣ Zoonotic- communicable disease transmitted from vertebrate animals to man under natural conditions e.g. rabies, plague, anthrax, salmonellas, hydatid cyst, bovine TB, and monkey pox.
- ▣ Man to vertebrate: zoonanthroposis e.g. TB
- ▣ Vertebrate to man: anthropozoonosis e.g. rabies

Component...

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- Epizootic (epidemic in animals) e.g. anthrax, brucellosis, rabies, Q-fever. encephalitis
- Enzootic (endemic in animals) : e.g. anthrax, rabies, brucellosis and endemic typhus

Two main modes of transmission:

1. Direct

- **Direct contact:** no intervening agency from reservoir/source to a susceptible host e.g. STD, AIDS and scabies.
- **Droplet infection:** direct projection of droplets spray of saliva or nasopharyngeal secretion 30-60cm.
- A type of particle implicated in the spread of airborne infection.
- Droplet nuclei are tiny particles (1–10 μm diameter) that represent the dried residue of droplets.
 - ▣ In to hosts nose or mouth by acts like sneezing coughing or talking e.g. common cold, diphtheria, TB, meningococcal meningitis. Risks are overcrowding and lack of ventilation.

Component...

135

- **Contact with soil:** tetanus, mycosis
- **Bite of animal:** rabies
- **Placental/vertical(**intergenerational**):** syphilis, AIDS, German measles, drugs.

2. Indirect

- **Vehicle:** water, food, milk, milk products, ice, blood, serum, plasma or other biological products.
- **Vector:** is an organism that transports an infectious agent to a susceptible host or to a suitable vehicle.
 - **Biological:** development or multiplication of disease causing organism before it is being transmitted to susceptible host.
 - **Mechanical:** is one in which the agent is directly infective to the host, and hence need not go through a period of multiplication or development.
 - Invertebrate: flies, mosquitoes, fleas, lice bugs, ticks, mites and Cyclops
 - Vertebrate: mice, rats and bats

Con't...

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- Man-arthropod: malaria
- Mammal or bird-arthropod: plague, Japanese encephalitis
- **Air (droplet nuclei):** TB, influenza, chicken pox, measles and Q-fever/dust=TB, streptococcal infection.
- **Fomites:** inanimate things other than food, water or milk e.g. cloths (soiled), towels, handkerchiefs, cups, spoons, toys, pencils, door handle, lavatory chains, syringes and surgical dressings diphtheria, typhoid, bacillary dysentery, eye and skin infections
- **Hands and fingers:**
 - ▣ Risks factors are lack of hygiene
 - ▣ Pathogens from skin, nose, feces are transmitted to susceptible host
 - e.g. typhoid, dysentery, infective hepatitis, intestinal worms, staphylococcal and streptococcal infections.

Component...

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5. Portal of entry

- Route through which infectious agent enters the host.
 - ▣ Gastrointestinal Tract (GIT) e.g. Typhoid fever, amoebiasis etc.
 - ▣ Respiratory Tract e.g. Tuberculosis, common cold
 - ▣ Skin and Mucous membrane e.g. STDS, Scabies, ring worm

6. Susceptible host

- This is a person who is highly likely to acquire infection when exposed to the agent
 - ▣ **Host-** a person or animal that harbors an infectious agent under natural conditions
 - ▣ **Obligate host-**the only host for specific organism e.g. man in typhoid
 - ▣ **Definitive or primary host-** one in which parasite attains maturity (man in ascaris) or passes its sexual stage (mosquito in malaria).
 - ▣ **Intermediate host-** one in which larval or asexual stage of parasite is spent (man in malaria)

Person to person transmission

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- Person to person transmission of an infectious agent is one of the main methods of disease spread in a community and is dependent on:
 - **Generation time**
 - This refers to the period between exposure/infection and the **maximum communicability** of the exposed host regardless of whether the disease is apparent or unapparent.
 - In case of apparent infections generation time may be equivalent to incubation period.
 - **Herd immunity**
 - This refers to a community resistance to spread of an infectious agent as a result of immunity gained by high proportion of individual members of the community
 - Though it may not be important to achieve 100% immunity, successful breakage of the chain of infection can be achieved if the immunity is close to 100%
 - **Secondary attack rate**
 - This is an important measure of spread of disease among contacts of an **index case**.
 - It has great use in epidemic situations

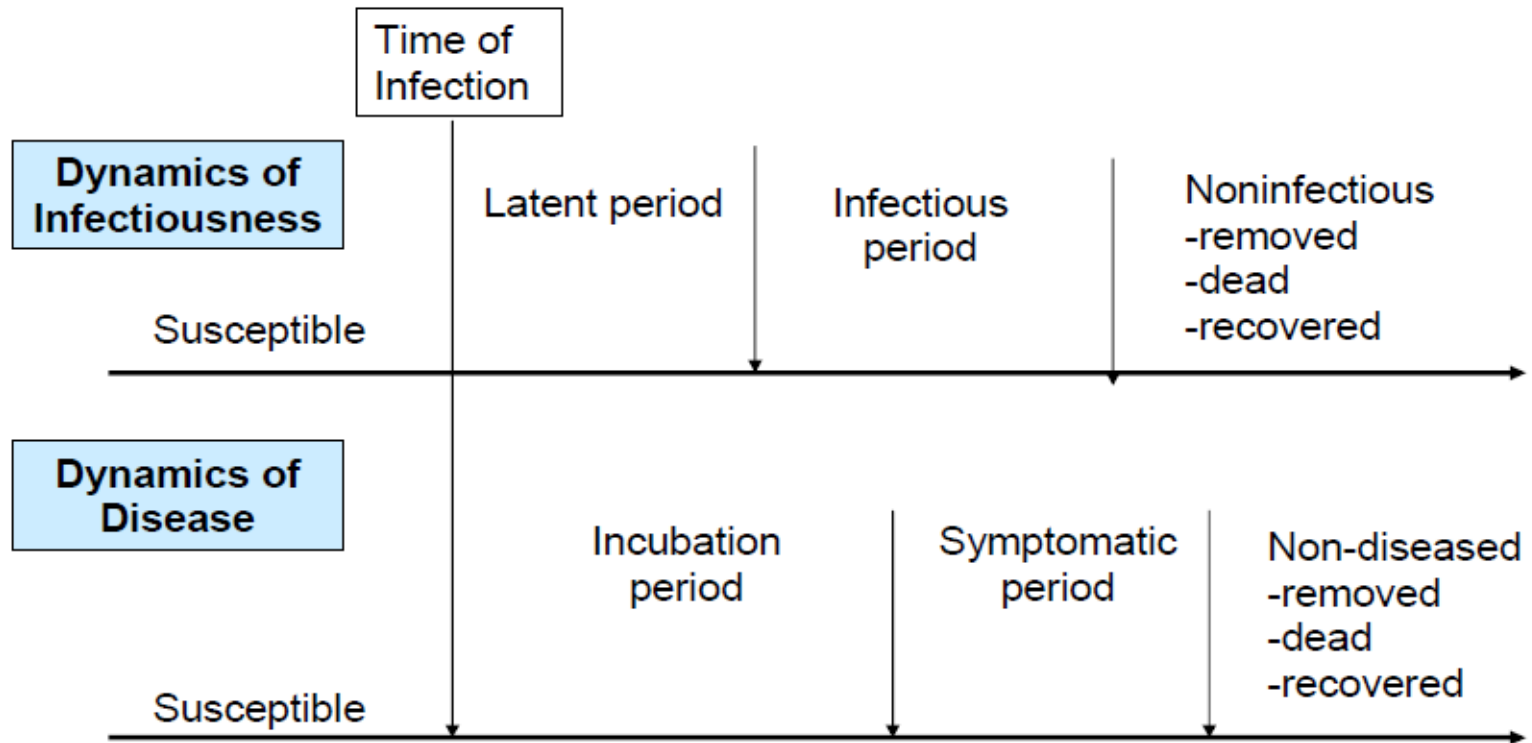
Time course of an infectious disease

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- The time lines of infection begin with the successful infection of the susceptible host by infectious agent
- **Dynamics of infection:** susceptible, latent period, infectious period and non infectious (removed, dead, recovered).
=> Time line for infection is important for the parasite/agent and public health
- **Dynamics of disease:** susceptible, incubation period, symptomatic period, non diseased (dead, removed, immune, carrier).
=> Time line for disease is important to infected person and physician/health worker

Time lines for Infection and Disease

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Time course of an infectious disease

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- ***Latent period***- the time interval between recovery and occurrence of relapse/recrudescence in clinical disease.
 - ▣ the time interval from infection to development of infectiousness
- ***Prelatent period***- time interval between infections (biological onset) to a point at which the infectious agent starts to shed
 - ▣ It can be first detected laboratorial
- ***Communicable period (Infectious period)***: -the period during which an infected host can transmit the infection to others
- ***Incubation period***- it includes the time interval between infection (biological onset) and the first clinical manifestation of the disease.

Incubation Period...

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- Is the time interval between **entry** and **development** of signs and symptoms of disease.
- It is the time interval between the entry of pathogenic micro organism and sufficient multiplication to cause pathology that will produce signs and symptoms of disease.
 - ▣ Incubation period (in non infectious diseases): is equivalent to induction + latent period.
 - ▣ Chicken pox- latent period is less than incubation period, therefore quarantine has no use
 - ▣ Malaria has incubation period of 14 days, stages infective to mosquito develop 10 days after incubation period.

Incubation...

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- HIV- short latent period. Two periods of latency:
 - ▣ Latent period for infectiousness
 - ▣ Latent period for seroconversion
 - Virulence, load and host response determine incubation period
- Short incubation period implies 2-3 days e.g. cholera, influenza
- Intermediate incubation period implies 10-21 days e.g. typhoid
- Long incubation periods imply several weeks- months/years e.g. leprosy, TB and rabies

Incubation...

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USE OF KNOWLEDGE OF IP

- ▣ Trace source of infection and contacts
- ▣ Determine the period of surveillance/ quarantine
- ▣ Prevention of disease by using immunoglobulin and antiserum
- ▣ Determine an epidemic as point source or propagated/ exponential
- ▣ Assess prognosis of an infectious disease.

◇ **PRIMARY CASE:** The individual who introduces the disease into the family or group under study. Not necessarily the first diagnosed case in a family or group.

◇ **Index case:** The first case in a family or other defined group to come to the attention of the investigator: propositus in genetics

Difficulties in infectious disease study

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- Identification of infectious source
- Identification of susceptible host
- Quantification of infectiousness and susceptibility
- Knowing strain of the parasite
- Defining and identifying contacts between infective and susceptible
- Contact rate efficacy: is change in contact due to intervention
- Exposure or behavior efficacy: is exposure change due to intervention
- Risk factors for susceptibility vs. risk factors for exposure
- Transmission probability: is conditional while incidence rate, and incidence proportion are unconditional

Carriers and their role in disease transmission

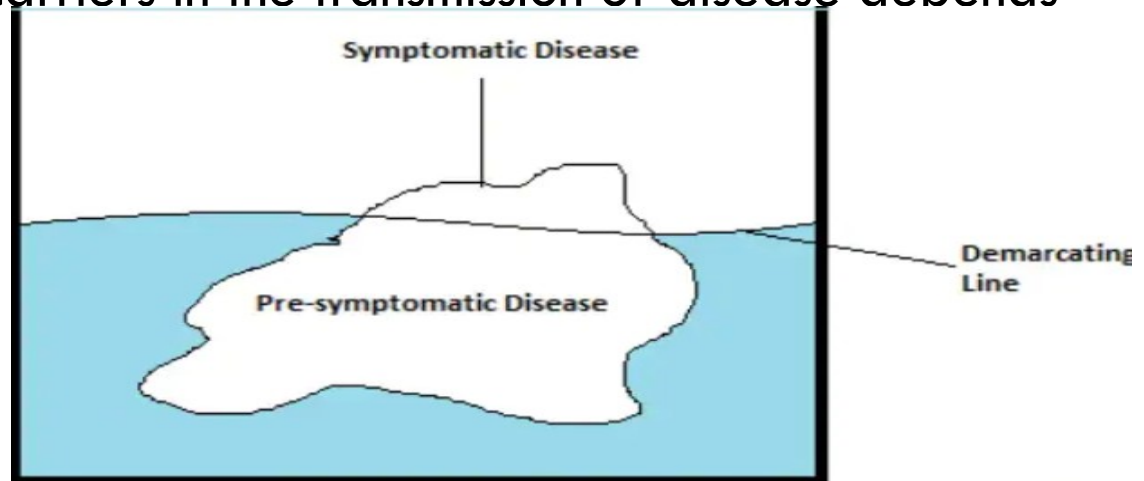
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- **Carrier State:** A condition in which a man or animal harbors and excretes infectious agent without **overt clinical signs and symptoms**
- Are **hidden reservoir** of infection and that they may outnumber actual cases.
- **Carrier** is a person without apparent disease who is nonetheless capable of transmitting the agent to others

Iceberg/ Hippopotamus effect

- The importance of carriers in the transmission of disease depends on their:

- Number
- Detectability,
- Mobility, and
- Chronicity



Carriers...

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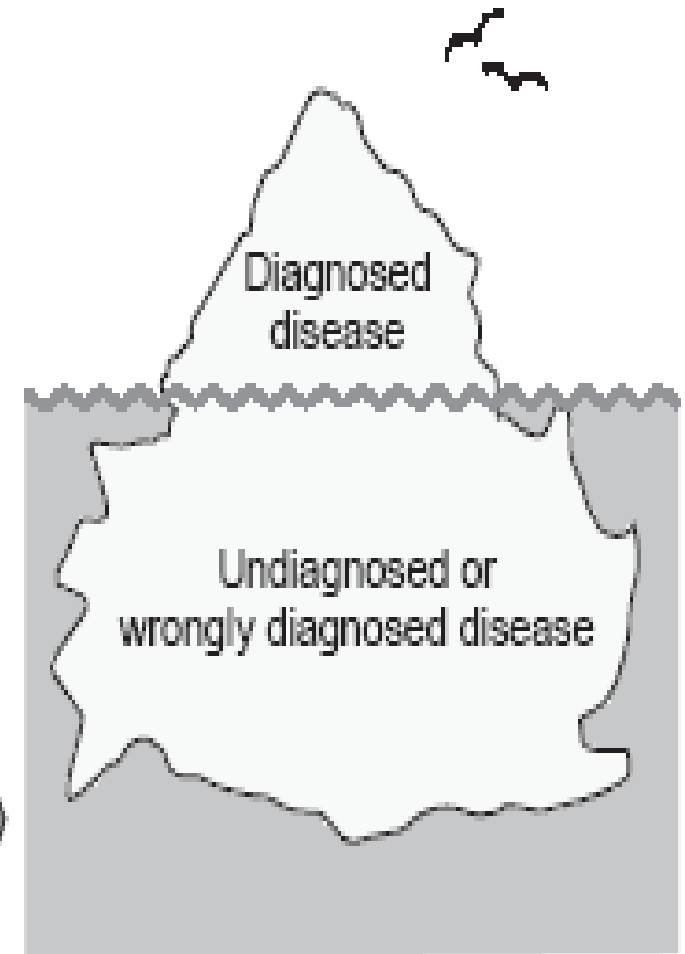
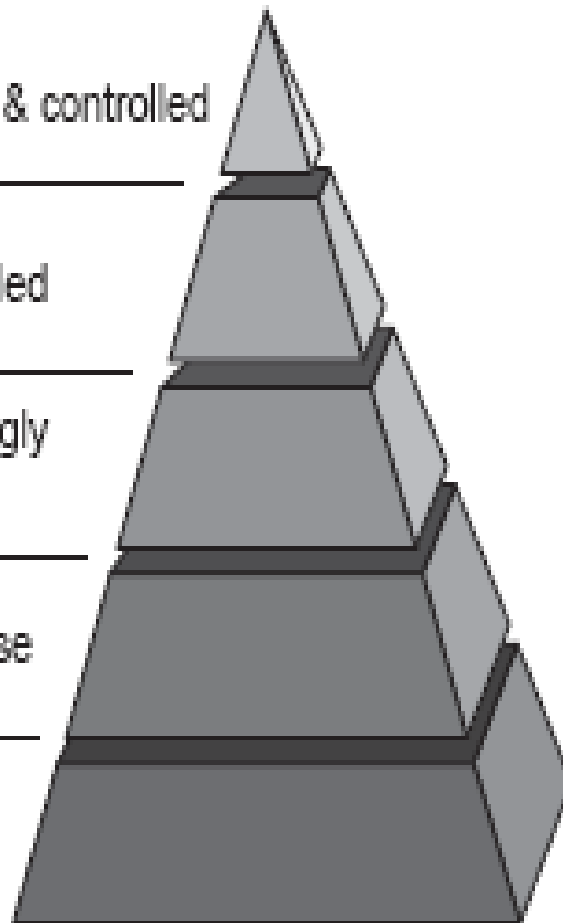
1 Diseased, diagnosed & controlled

2 Diagnosed, uncontrolled

3 Undiagnosed or wrongly diagnosed disease

4 Risk factors for disease

5 Free of risk factors



CARRIER ...

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- Epidemiologically, carriers are more important than overt clinical cases because identification may require sophisticated investigation but important to control transmission e.g. AIDS
- Carriers may be classified based on **type, duration, and portal** of exit. **Carriers may be:**
- **Healthy / Asymptomatic / Carrier:** in this type of carrier state infection remains inapparent, i.e. no signs and / or symptoms suggesting clinical disease are manifest
- Transmitting infection without ever showing signs of the disease.
e.g. meningococcal meningitis ,typhoid fever, hepatitis B virus.
- **Incubatory carriers** - transmitting infection by shedding the agent before the onset of clinical manifestations or
 - ▣ These are infected individuals who excrete the pathogen during the incubation period, i.e. before the onset of clinical disease. e.g. Measles, mumps, hepatitis

Carrier...

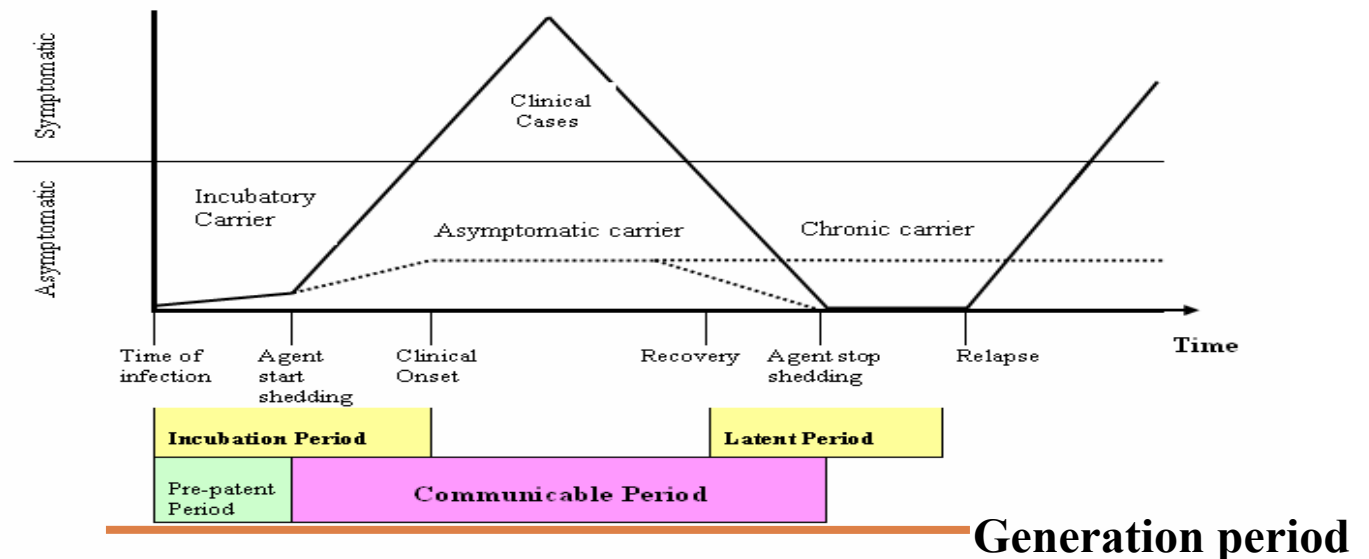
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- **Convalescent carriers** - These are people who continue to harbor and excrete the infectious agent after **recovering** from the illness
 - ▣ Transmitting infection after the time of recovery from the disease
e.g. Salmonella typhi, hepatitis B, .
- **Chronic carriers** - In this type, the carrier state persists for a long period of time or indefinitely.
 - ▣ Shed the agent for a long period of time, or even indefinitely. e.g. Salmonella typhi, hepatitis B virus.

Carrier...

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- **Prepatent period:** the time interval from infection to becoming infectious (shedding of the agent).
- **Infectious period:** the time during which an infected host could infect another host or vector. (**Communicable period**)
- **Incubation period:** the time interval from infection to symptomatic disease.





Thank You!!!